A one-dimensional area law and a matrix product state approximation from a global spectral gap: the collar-free route

Prepared for Sirui Lu[1]

[1] *Max Planck Institute of Quantum Optics, Garching*

(Dated: October 16, 2026)

We prove an entanglement area law and a polynomial matrix product state approximation for the unique gapped ground state of any finite-range one-dimensional Hamiltonian, specialising the two-dimensional arguments of *A two-dimensional area law from a global spectral gap* [1] and *Polynomial PEPS approximation of gapped square-grid ground states* [2] to a spin chain. Two structural simplifications occur on a line. First, the area-law proof collapses to three elementary steps: the marginal tail bound and the buffered conditional-entropy discount of [1, Section 2], combined with the Araki–Lieb inequality, directly yield $S(X) \le C(q,R)\,(1+J/\Delta)$ for every interval X. The reason is geometric: a shell around an interval is crossed by a bounded number of interactions, so the conditional-entropy discount costs a constant rather than a boundary-scaled error, and the buffer whose entropy must be controlled contains only a constant number of sites. The sub-volume recursion, positive quasi-local constraints, replica scanner, amplification, and two-family geometry of [1] drop out entirely. Second, the tensor-network construction reduces from a dyadic hierarchy to a linear left-to-right *sweep*. The encoded holes, adaptive patch constraints, angular information estimates, and row–column routing of [2] disappear; what remains are Uhlmann-type birth, death, and exchange moves, an information reset via a two-sided Gaussian spectral filter, dimension-free Gaussian density compression, and column extraction. The output is a matrix product state of bond dimension CL^c within vector error L^{-1} of the ground state. We also record that in one dimension the marginal tail bound makes the matrix product state theorem a two-line corollary of the area law by sequential Schmidt truncation. Every statement in this note is written twice: first in words, then precisely. The area-law proof uses none of the techniques of Hastings or of Arad–Kitaev–Landau–Vazirani; the tensor-network protocol uses a Lieb–Robinson bound in a single lemma, and we indicate how to avoid it at the price of a quasi-polynomial bond dimension.

I. OVERVIEW

In words. In a gapped quantum lattice system, entanglement across a region is generated along its boundary. In two dimensions, a region of linear size r has a boundary of $O(r)$ sites: every variational estimate across the boundary incurs an $O(r)$ error, and any buffer introduced to isolate the region is an annulus whose own entanglement is unknown. The multi-scale replica, scanner, and amplification apparatus of [1] exists to break this circularity. In one dimension, the boundary of an interval is at most two points. Any variational operation across the boundary disturbs a bounded number of interactions and costs $O(1)$ energy, while a buffer of constant width has, trivially, constant entropy. The circularity never arises: a marginal tail bound, a buffered conditional-entropy discount, and the Araki–Lieb inequality establish the area law in full. This is the content of Part I.

Constructing a global tensor-network approximation requires more than entropy bounds alone (the papers explain why: an entropy bound on each cut does not give a global vector approximation). The two-dimensional construction of [2] therefore distributes the ground state among "parties" attached to dyadic squares via local changes of ownership controlled by mutual information, and then compresses the resulting distributed density operator into a tensor network. On a line, a dyadic hierarchy is the wrong geometry: a binary tree of intervals has cutwidth $\log L$ across the middle of the chain, which turns a polynomial link dimension into a quasi-polynomial bond dimension. The right geometry is a left-to-right sweep with a moving "remainder" party, which keeps the cutwidth bounded. The only topological obstruction during the sweep is dissolving a narrow remainder band wedged between two distinct neighbouring owners; this is resolved by the same sequence of Uhlmann birth, death, and two-sheet exchange moves used along block edges in [2]. This is Part II. We also show that, in one dimension, the marginal tail bound of Part I makes the matrix product state theorem an immediate corollary of the area law by sequential Schmidt truncation; the sweep protocol is the one-dimensional counterpart of the two-dimensional construction, and it is the mechanism that generalises to higher dimensions.

Table I summarises how the two-dimensional architecture specialises to a line. Readers interested solely in the one-dimensional area law need only read Sections II–V (about four pages).

Precisely. Throughout, $q \ge 2$ and $R \ge 1$ are integers and $J, \Delta > 0$. A finite set $\Lambda \subset \mathbb{Z}$ carries the induced path graph (with edges between consecutive integers that both lie in Λ) and graph distance d_Λ. Each site carries $\mathbb{C}^q$ and $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$. An admissible Hamiltonian is

$$H = \sum_{\substack{\varnothing \ne X \subseteq \Lambda \\ \operatorname{diam}_{d_\Lambda} X \le R}} h_X, \qquad h_X = h_X^*, \qquad \|h_X\| \le J, \quad (1)$$

with one summand per support and zero summands allowed. The standing spectral hypothesis is that H has a unique normalised ground vector Ω (up to phase), with ground energy E_0 and spectral gap Δ:

$$H - E_0 \mathrm{I} \;\ge\; \Delta\big(\mathrm{I} - |\Omega\rangle\langle\Omega|\big). \quad (2)$$

Two-dimensional ingredient	Specialisation to one dimension
Marginal tail bound ([1], Lemma 2.1)	Kept verbatim; the cut parameter $\mathcal{B}$ is a constant (Section III).
Buffered conditional-entropy discount ([1], Lemma 2.2)	Kept; the buffer has *constant* width and the error is a constant (Section IV).
Sub-volume recursion, exponent $e_0 < 1$	Unnecessary: the Araki–Lieb inequality closes the argument directly (Section V). Sketched for comparison in Section VI.
Positive quasi-local constraints, truncation	Unnecessary for the area law; replaced by the Gaussian filter reset in the MPS protocol.
Conditional estimates, replicas, transport, comparators, scanner	Unnecessary.
Amplification by random constraint products	Unnecessary; Section VI explains how it would apply on a line.
Two families of regions	Trivial on a half-line; Section VI.
Coarse collar and angular information ([2], Section 3)	Replaced by a direct consequence of the area law and information reset (Lemma VIII.5).
Information reset by Gaussian filter ([2], Lemma 3.3)	Kept with one-dimensional Lieb–Robinson locality (Section VIII B).
Adaptive patch constraints, encoded holes ([2], Sections 4, 6.1)	Unnecessary: no three owners meet on a line, and logarithmic blocks have polynomial Hilbert-space dimension.
Births, deaths, exchanges ([2], Lemmas 6.6–6.8)	Kept verbatim (Section IX).
Dyadic hierarchy with edge bands and point treatments ([2], Section 7)	Replaced by a left-to-right linear sweep (Section X).
Density compression, column extraction ([2], Sections 5, 8)	Kept verbatim (Section XI).
Row–column routing with constant congestion	Replaced by direct placement along the chain (Section XII).

TABLE I. Correspondence between the two-dimensional constructions of [1, 2] and their one-dimensional specialisation in this note: what survives, what simplifies, and what drops out.

For $A \subseteq \Lambda$, let $\rho_A = \mathrm{Tr}_{\Lambda\setminus A}\,|\Omega\rangle\langle\Omega|$ be the reduced density operator and $S(A) = S_\Omega(A) = -\operatorname{Tr}\rho_A \log \rho_A$ its von Neumann entropy (with natural logarithms and $0\log 0 = 0$). For disjoint $A, B \subseteq \Lambda$ (writing AB for $A\cup B$), the conditional entropy and mutual information are $S(A|B) = S(AB) - S(B)$ and $I(A:B) = S(A) + S(B) - S(AB)$. The edge boundary of A in Λ is $\partial_\Lambda A = \{\{x, x+1\} \subseteq \Lambda : \text{ exactly one of } x, x+1 \text{ lies in } A\}$. Constants C, c depend only on q, R, J, Δ unless stated otherwise and may change from line to line; where a specific dependence is tracked, it is stated explicitly. The two main theorems of this note are as follows.

Theorem I.1 (One-dimensional area law)**.** *There is $C_{\mathrm{AL}} = C_{\mathrm{AL}}(q, R, J, \Delta)$, with*

$$C_{\mathrm{AL}} \le C_0(q, R)\left(1 + \frac{J}{\Delta}\right),$$

such that for every finite $\Lambda \subset \mathbb{Z}$, every admissible Hamiltonian satisfying (2), and every $A \subseteq \Lambda$,

$$S_\Omega(A) \le C_{\mathrm{AL}}\,|\partial_\Lambda A|.$$

Theorem I.2 (One-dimensional matrix product state approximation)**.** *Fix q, R, J, Δ. There are $C, c > 0$ depending only on these parameters such that for every $L \ge 2$ and every admissible Hamiltonian on $\Lambda = \{1, \dots, L\}$ satisfying (2), there is a nonzero matrix product state Φ on Λ with open boundary conditions and bond dimension at most CL^c such that*

$$\min_{\theta\in\mathbb{R}} \left\| \frac{\Phi}{\|\Phi\|} - \mathrm{e}^{i\theta}\Omega \right\| \le L^{-1}.$$

The constants never depend on Λ, on A, or on the specific Hamiltonian. No spectral gap for restricted sub-Hamiltonians, no frustration-freeness, and no translation invariance are assumed.

a. Formalization. Theorems I.1 and I.2, with explicit constants, are formalized in Lean 4 in the TNLean modules `TNLean/MPS/AreaLaw/` (blueprint chapter "The one-dimensional area law and matrix product state approximation"), stated directly on the chain of L sites. The counting constants N_R, ℓ_R, $\mathcal{B}_R$ and the shell width R are those printed here. The buffered discount is run at the energy threshold $\Delta/2$ in place of $\Delta/4$, the shell count omits the last entry of (14), and the constant $1 + 2C_E/\Delta$ of (16) is replaced by $4096\,\mathrm{e}\,\vartheta\,\mathcal{B}_R/m + m\log 2$, which changes C_{int} and C_0 accordingly; these deviations are recorded in `docs/paper-gaps/tnlean_one_dimensional_discount_constants.tex`. Theorem I.1 is formalized for connected domains, the chains, without the component reduction of Section II (`docs/paper-gaps/tnlean_one_dimensional_connected_domain_scope.tex`); Theorem I.2 is formalized by the two-line route of Section VII only, with Proposition VII.1 in its threshold form (projection onto the

eigenvalues at least a threshold rather than onto the D largest; `docs/paper-gaps/tnlean_one_dimensional_threshold_truncation.tex`), and the protocol route of Part II is not formalized.

Part I
The one-dimensional area law

II. SETTING, REDUCTIONS, AND COUNTING

In words. To bound the entanglement entropy $S_\Omega(A)$ of an arbitrary subset $A \subseteq \Lambda$ in terms of its edge boundary $|\partial_\Lambda A|$, three elementary reductions allow us to restrict attention to a single interval X inside a connected chain. First, if the domain Λ has several connected components, every interaction support has finite graph diameter and therefore lies inside a single component; the Hamiltonian decomposes as a sum over components, the unique ground vector factors as a tensor product, and both the entanglement entropy and the edge boundary are additive across components. Thus we may take Λ to be a single interval. Second, any subset A of an interval is a disjoint union of maximal subintervals, each contributing at least one distinct boundary edge, so subadditivity of the entropy reduces Theorem I.1 to the case where A is an interval. Third, an interval has at most two boundary cut edges, and any range-R interaction crossing a cut edge is supported within a window of $2R$ consecutive sites centred at that cut. Both the number of interactions crossing the boundary of an interval and the logarithm of their joint Hilbert-space dimension are therefore bounded by constants depending only on R and q. This boundary counting is the entire geometric input of the one-dimensional area law.

Precisely. For an interval $X \subseteq \Lambda$ and a support Y with $\mathrm{diam}_{d_\Lambda} Y \le R$, we say that Y is *split* by X if $Y \cap X \neq \varnothing$ and $Y \setminus X \neq \varnothing$. Since Y is contained in $R+1$ consecutive integers and contains a site on each side of one of the two cut edges of X, it lies entirely in one of the two windows of $2R$ consecutive sites centred at a cut edge. Hence

$$\begin{aligned} &\#\{Y:\ Y \text{ split by } X\} \le 2 \cdot 2^{2R} =: N_R, \\ &\dim \mathcal{H}_Y \le q^{R+1}, \qquad \ell_R := \log(e q^{R+1}). \end{aligned} \tag{3}$$

To separate the action of a split interaction h_Y across the cut between X and $\Lambda \setminus X$, we expand h_Y in the matrix units E_{ab} of $\mathcal{H}_{Y\cap X}$ as

$$h_Y = \sum_{\alpha=1}^{N_Y} c_\alpha \otimes d_\alpha, \qquad N_Y \le q^{2(R+1)}, \qquad \|c_\alpha\|\,\|d_\alpha\| \le J, \tag{4}$$

with c_α supported in X and d_α in $\Lambda\setminus X$ (each matrix unit E_{ab} on $\mathcal{H}_{Y\cap X}$ has operator norm one, and the coefficient operator $d_\alpha = \mathrm{Tr}_{Y\cap X}[(E_{ba}\otimes \mathrm{I})h_Y]$ has norm at most $\|h_Y\| \le J$). Put

$$\mathcal{B}_R := 1 + N_R\, \ell_R^2. \tag{5}$$

Lemma II.1 (Reductions)**.** *It suffices to prove Theorem I.1 for Λ an interval and A an interval with $\partial_\Lambda A \neq \varnothing$, with a constant C_{int} bounding $S_\Omega(A)$; then Theorem I.1 holds with $C_{\mathrm{AL}} = C_{\mathrm{int}}$.*

Proof. Let $\Lambda_1, \dots, \Lambda_k$ be the connected components of Λ (maximal intervals). Every support of finite graph diameter lies in a single component, so $H = \sum_i H_i$ with H_i acting on $\mathcal{H}_{\Lambda_i}$. The ground eigenspace of H is the tensor product of the ground eigenspaces of the H_i; since the former is one-dimensional, so is each of the latter, hence $\Omega = \bigotimes_i \Omega_i$ and $S_\Omega(A) = \sum_i S_{\Omega_i}(A \cap \Lambda_i)$, while $\partial_\Lambda A = \bigsqcup_i \partial_{\Lambda_i}(A \cap \Lambda_i)$. Each H_i inherits (2) with the same gap Δ (by restricting the inequality to vectors of the form $\xi \otimes \bigotimes_{j\neq i} \Omega_j$). We may therefore assume that Λ is an interval. Write $A = \bigsqcup_{a=1}^{m} A_a$ as a union of maximal subintervals. If some $A_a = \Lambda$, then $A = \Lambda$ and $S(A) = 0$. Otherwise each A_a has at least one boundary edge, and distinct A_a have disjoint boundary edges (any two maximal subintervals are separated by at least one site outside A, so their outer cut edges are distinct). Thus $m \le |\partial_\Lambda A|$, and subadditivity yields $S(A) \le \sum_a S(A_a) \le C_{\mathrm{int}} |\partial_\Lambda A|$. □

From now on $\Lambda = \{1, \dots, N\}$ and $X \subseteq \Lambda$ is an interval. For an integer $j \ge 0$, write $X^{+j} = \{x \in \Lambda : \mathrm{dist}(x, X) \le j\}$, which is again an interval.

III. THE MARGINAL TAIL BOUND

In words. Fix an interval X and write $\Omega = \sum_j \sqrt{p_j}\, u_j \otimes w_j$ for its Schmidt decomposition across X and $\Lambda \setminus X$. Both the variational filter norms in Section IV and the Schmidt truncation in Part II require controlling moments $\mathrm{Tr}\,\rho_X^{1-u}$ of the reduced density operator in terms of the von Neumann entropy $S(X)$ alone. Viewing the entanglement spectrum as the probability distribution of the *surprisal* $K = -\log p_j$ drawn with probability p_j, whose mean is $\mathbb{E}K = S(X)$, this amounts to showing that K is concentrated around $S(X)$. Lemma III.1 shows that the global spectral gap forces K to have sub-Gaussian moments near $u = 0$ and exponential tails around $S(X)$, with a width determined solely by q, R, and J/Δ, regardless of how large $S(X)$ may be.

The physical mechanism is a non-unitary marginal tilt. Multiply Ω on X by $F = \rho_X^{-u/2}$ and renormalise: the resulting trial vector $\Theta = F\Omega/\sqrt{M(u)}$ has Schmidt probabilities $p_j^{1-u}/M(u)$, where $M(u) = \sum_j p_j^{1-u} = \mathbb{E}\,\mathrm{e}^{uK}$ is the moment generating function of K. Because $\Theta \propto F\Omega$, it is an exact eigenvector of the conjugated Hamiltonian FHF^{-1} with energy E_0, so its excitation energy $\langle \Theta, (H - E_0)\Theta\rangle$ equals $\langle \Theta, (H - FHF^{-1})\Theta\rangle$. Interactions supported inside $\Lambda\setminus X$ commute with F and cancel; interactions supported inside X also cancel in expectation because F commutes with the tilted marginal $\sigma = \mathrm{Tr}_{\Lambda\setminus X}\,|\Theta\rangle\langle\Theta|$. Only the at most N_R interactions split by the boundary of X contribute.

To see why each split interaction h contributes quadratically in u rather than linearly, embed the conjugation into the analytic family $f(z) = \langle \Theta, \sigma^z h \sigma^{-z}\Theta\rangle$. On the imaginary

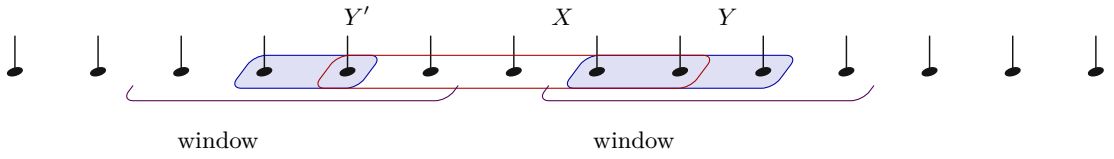


FIG. 1. An interval X of the chain (drawn for $R = 2$), the two windows of $2R$ sites centred on its cut edges, and two interaction supports Y' and Y split by X, one in each window. Every split support lies in a window, which is why their number is bounded by N_R.

axis σ^{iy} is unitary, so $|f(iy)| \le J$; on the strip $|\Re z| \le \frac{1}{4}$, a matrix-unit expansion bounds $|f(z)|$ by $2Jq^{2(R+1)}$, and Hadamard's three-lines theorem compresses the dimensional factor into a strip of width $\delta_R \propto \ell_R^{-1}$ on which $|f(z)| \le \mathrm{e}J$. Crucially, Hermiticity of h gives the reflection symmetry $f(-\bar z) = \overline{f(z)}$, which makes all odd Taylor coefficients at the origin purely imaginary and eliminates the linear term of $\Re(f(x) - f(0))$. Cauchy's estimate therefore bounds the excitation energy by $O(J\mathcal{B}_R u^2)$. The spectral gap converts this quadratic energy into a quadratic lower bound on the ground-state overlap $|\langle\Omega, \Theta\rangle|^2 = M(u/2)^2/M(u) \ge 1 - O(\vartheta\mathcal{B}_R u^2)$, or $\log M(u) \le 2\log M(u/2) + Cu^2$. Iterating this scale-halving relation down to $u \to 0$, where $\log M(u) = uS(X) + o(u)$, yields the sub-Gaussian bound $\log M(u) \le uS(X) + Cu^2$, and Markov's inequality gives the exponential tail bound.

In two dimensions the tail width scales as the square root of the boundary length; in one dimension it is an $O(1)$ constant, which underpins both parts of this note.

Precisely. Let $X \subseteq \Lambda$ be an interval, let $\rho = \rho_X$ have positive eigenvalues p_j, let K be the random variable taking the value $-\log p_j$ with probability p_j so that $\mathbb{E}K = S(X)$, and put

$$\vartheta = \frac{J}{\Delta}, \qquad u_0 = \frac{1}{32\sqrt{(1+\vartheta)\mathcal{B}_R}}. \tag{6}$$

Lemma III.1 (Marginal tail bound)**.** *For every interval $X \subseteq \Lambda$ and every real u with $|u| \le u_0$,*

$$\log \operatorname{Tr} \rho_X^{1-u} = \log \mathbb{E}\, \mathrm{e}^{uK} \le u\, S(X) + 512\, \mathrm{e}\, \vartheta\, \mathcal{B}_R\, u^2, \tag{7}$$

where ρ_X^{1-u} is taken to be zero on the kernel of ρ_X. Consequently, for every $w \ge 0$,

$$\Pr\{|K - S(X)| > w\} \le 2\mathrm{e}^{\mathrm{e}/2} \exp(-u_0 w). \tag{8}$$

Proof. Put $M(u) = \sum_j p_j^{1-u}$ and $L(u) = \log M(u)$; note that $L(0) = 0$ and $L'(0) = S(X)$. Let P be the support projection of ρ, set $F = \rho^{-u/2}$ on $P\mathcal{H}_X$ and $F = \mathrm{I}$ on the kernel, and define

$$\Theta = \frac{F\Omega}{\sqrt{M(u)}}, \qquad \sigma = \operatorname{Tr}_{\Lambda\setminus X} |\Theta\rangle\langle\Theta| = \frac{\rho^{1-u}}{M(u)}.$$

Since F is invertible on $\mathcal{H}_\Lambda$, we have $FHF^{-1}\Theta = E_0\Theta$ and

$$\langle\Theta, (H - E_0)\Theta\rangle = \sum_Y \big(\langle\Theta, h_Y\Theta\rangle - \langle\Theta, Fh_YF^{-1}\Theta\rangle\big). \tag{9}$$

A term h_Y supported in $\Lambda \setminus X$ commutes with F and contributes zero. A term supported in X also contributes zero because $\operatorname{Tr}(\sigma F h_Y F^{-1}) = \operatorname{Tr}(\sigma h_Y)$ by $[F, \sigma] = 0$ and cyclicity of the trace. Thus only terms split by X remain, and there are at most N_R of them by (3).

To bound each split term $h = h_Y$ analytic-function-theoretically, define for $z \in \mathbb{C}$ the operator $\sigma^z = P \exp(z \log(\sigma|_{P\mathcal{H}_X}))P$ and the entire function $f(z) = \langle\Theta, \sigma^z h \sigma^{-z}\Theta\rangle$, which satisfies $f(0) = \langle\Theta, h\Theta\rangle$ since Θ lies in the support of σ. At $z_u = -u/(2(1-u))$ one has, on the support, $\sigma^{z_u} = M(u)^{-z_u}F$ and $\sigma^{-z_u} = M(u)^{z_u}F^{-1}$, so $f(z_u) = \langle\Theta, FhF^{-1}\Theta\rangle$. On the imaginary axis the powers σ^{iy} are unitary on the support, giving $|f(iy)| \le J$. To bound $f(z)$ on the strip $|\Re z| \le \frac{1}{4}$, expand h in matrix units as in (4) and write $\Theta = \sum_j \sqrt{s_j}\, e_j \otimes f_j$ for its Schmidt decomposition across X:

$$f(z) = \sum_\alpha \sum_{j,k} s_j^{1/2+z} s_k^{1/2-z}\, \langle e_j, c_\alpha e_k\rangle\langle f_j, d_\alpha f_k\rangle.$$

For $|\Re z| \le \frac{1}{4}$, the weighted arithmetic–geometric mean inequality gives $|s_j^{1/2+z} s_k^{1/2-z}| \le s_j + s_k$, and for any operators c, d with matrix elements c_{jk}, d_{jk} the row and column Cauchy–Schwarz inequalities yield

$$\sum_{j,k} (s_j + s_k)\, |c_{jk}|\, |d_{jk}| \le 2\|c\|\, \|d\|, \tag{10}$$

because $\sum_k |c_{jk}||d_{jk}| \le \|c\|\|d\|$ for each row j and likewise for each column k, while $\sum_j s_j = 1$. Hence $|f(z)| \le 2Jq^{2(R+1)}$ on $|\Re z| \le \frac{1}{4}$. Applying Hadamard's three-lines theorem on each of the strips $0 \le \pm\Re z \le \frac{1}{4}$ gives $\log|f(x+iy)| \le \log J + 4|x| \log(2q^{2(R+1)}) \le \log J + 8|x|\, \ell_R$, so that

$$|f(z)| \le \mathrm{e}J \qquad \text{for } |\Re z| \le \delta_R := \frac{1}{8\ell_R}. \tag{11}$$

Since $h = h^*$ and $(\sigma^z)^* = \sigma^{\bar z}$, we have $f(-\bar z) = \overline{f(z)}$. In the power-series expansion $f(z) = \sum_n c_n z^n$, the coefficient c_n is therefore real for even n and purely imaginary for odd n. For real x, the linear term in the real part drops out, leaving $\Re(f(x) - f(0)) = \sum_{n \ge 2 \text{ even}} c_n x^n$, and Cauchy's estimate $|c_n| \le \mathrm{e}J\delta_R^{-n}$ on the disc of radius δ_R from (11) gives, for $|x| \le \delta_R/2$,

$$|\Re(f(x) - f(0))| \le 2\mathrm{e}J\, x^2 \delta_R^{-2} = 128\, \mathrm{e}J\, \ell_R^2\, x^2. \tag{12}$$

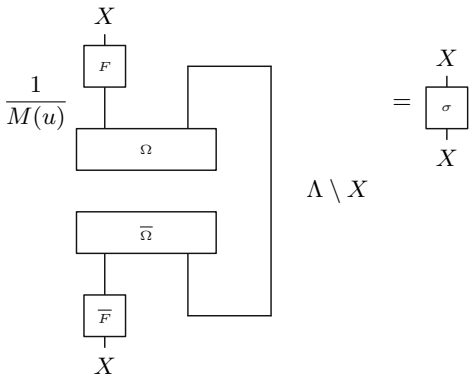


FIG. 2. The tilt as a tensor diagram. Multiplying Ω on the interval X by $F = \rho_X^{-u/2}$ and tracing out the rest of the chain gives $\sigma = \rho_X^{1-u}/M(u)$. Only interactions crossing the cuts of X fail to commute with F, so the tilted state has excitation energy of order u^2.

For $|u| \le u_0$ we have $|u| \le \frac{1}{32}$, $|z_u| \le |u|$, and $\ell_R \le \sqrt{\mathcal{B}_R}$, so $|z_u| \le \delta_R/2$. Since the left-hand side of (9) is real, applying (12) at $x = z_u$ to each of the at most N_R split terms yields

$$0 \le \langle \Theta, (H - E_0)\Theta \rangle \le 128\,\mathrm{e} J\, N_R \ell_R^2\, u^2 \le 128\,\mathrm{e} J\, \mathcal{B}_R\, u^2. \tag{13}$$

The spectral gap (2) converts this energy bound into the overlap estimate $1 - |\langle \Omega, \Theta \rangle|^2 \le 128\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2 \le \frac{1}{2}$. Since $\langle \Omega, \Theta \rangle = M(u/2)/\sqrt{M(u)}$, we obtain

$$\frac{M(u/2)^2}{M(u)} \ge 1 - 128\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2, \qquad \text{i.e.}$$
$$L(u) \le 2L(u/2) + 256\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2,$$

using $-\log(1-v) \le 2v$ for $0 \le v \le \frac{1}{2}$. Every halving $u/2^j$ remains in $[-u_0, u_0]$, so induction over j gives

$$L(u) \le 2^j L(u/2^j) + 256\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2 \sum_{i<j} 2^{-i}$$
$$\xrightarrow[j\to\infty]{} uS(X) + 512\,\mathrm{e}\,\vartheta\,\mathcal{B}_R u^2,$$

which proves (7). For (8), apply Markov's inequality to $\mathrm{e}^{\pm u_0(K - S(X))}$: at $u = \pm u_0$, the quadratic term in (7) is $512\mathrm{e}\vartheta\mathcal{B}_R/(1024(1+\vartheta)\mathcal{B}_R) \le \mathrm{e}/2$, so each one-sided tail is bounded by $\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w}$. □

Remark III.2 (Rényi entropies near one)**.** For $0 < u \le u_0$ and $\alpha = 1 - u$, (7) reads $S_\alpha(X) = \frac{1}{u}\log \mathrm{Tr}\,\rho_X^\alpha \le S(X) + 512\mathrm{e}\vartheta\mathcal{B}_R(1-\alpha)$. Since $S_\alpha \ge S$ for $\alpha < 1$, all Rényi entropies of order $\alpha \in [1-u_0, 1]$ of an interval lie within $16\mathrm{e}\sqrt{(1+\vartheta)\mathcal{B}_R}$ of its von Neumann entropy. This is used in Section VII.

IV. THE BUFFERED DISCOUNT WITH A CONSTANT-WIDTH BUFFER

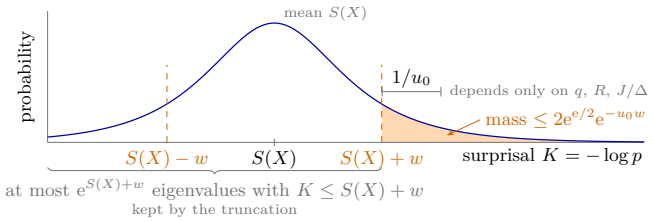


FIG. 3. Schematic law of the surprisal. The tails decay exponentially at rate u_0; the mass beyond $S(X) + w$ is at most $2e^{e/2}e^{-u_0 w}$, and at most $e^{S(X)+w}$ eigenvalues lie to the left of that line. Both facts are reused for the sequential truncation in Part II.

In words. This section establishes the core variational estimate of Part I. Conditioning an interval X on a buffer T of a constant number of nearby sites halves its entropy, up to an additive constant:

$$S(X \,|\, T) \;\le\; \tfrac{1}{2}\, S(X) + C.$$

To keep the excitation energy of the filtered vector below the gap Δ, the filtering is distributed across m concentric shells $X_j = X^{+jD}$ $(j = 1, \dots, m)$, each wider than the previous one by a step $D = R$, with flanking buffers $T_j = X_j \setminus X$ and small per-shell weight $a_j = 2/m$ summing to $\sum_j a_j = 2$. Choosing m of order $1 + J/\Delta$ keeps the total buffer $T = T_m$ of *constant* width mD on each side of X.

Specifically, we apply to Ω a product $K = L_m \cdots L_1$ of positive operators L_j acting on X_j, each normalised by $\mathrm{Tr}\, L_j^{2/a_j} = 1$, and maximise the norm $N = \|K\Omega\|$. Two complementary bounds on the maximum N yield the discount.

Upper bound on N (lower bound on $-2\log N$). Computing the excitation energy of the normalised filtered vector $\phi = K\Omega/N$ requires controlling the conjugation of each Hamiltonian term by all m filters. At a maximiser of N, however, unitary stationarity implies that each filter L_j commutes with the reduced density operator ρ_j of ϕ on its own shell X_j. This commutation gives a *stripping identity*: any operator supported inside X_k is unchanged in expectation when conjugated by the outer filter L_k. Because consecutive shells are separated by $D \ge R$, any range-R interaction h_Y is split by at most one shell X_j; all outer filters ($k > j$) contain Y and strip away, while all inner filters ($k < j$) are disjoint from Y and commute past h_Y. Thus h_Y is conjugated by L_j alone. Diagonal stationarity clips the log-eigenvalue differences of L_j to $a_j/2$ times those of ρ_j, so conjugation by L_j changes the expectation of h_Y by at most $O(Ja_j^2)$. Summing over the m shells and the at most N_R interactions split by each shell bounds the excitation energy of ϕ by $O(J\sum_j a_j^2) = O(J/m)$. The spectral gap then ensures a ground-state overlap $|\langle \Omega, \phi \rangle|^2 \ge 1 - C/(m\Delta)$, and the same holds for every prefix optimiser. Using this overlap to peel the filters L_j off one at a time, and bounding each peeled factor by $\|L_j\Omega\| \le \exp(-a_j S(X_j)/2 + Ca_j^2)$ via Schatten–Hölder and the marginal tail bound (Lemma III.1), gives $-2\log N \ge \sum_j a_j S(X_j) - C$.

Lower bound on N (upper bound on $-2\log N$). Conversely, we test the variational maximum on trial filters that "pin" a single Schmidt vector u_i of $\Omega = \sum_i \sqrt{p_i}\, u_i \otimes w_i$ across

X: namely $L_j = |u_i\rangle\langle u_i| \otimes M_j$, where M_j acts only on the flank T_j. This decouples X and produces $\sqrt{p_i}\,u_i$ tensored with $M_m \cdots M_1 w_i$. Because the flank filters M_j act on nested sets and need not commute, we relate the norm of their product to the sum of single-flank entropies via three-lines interpolation on $M_m^z \cdots M_1^z w_i$, finding that M_j can be chosen so that $\|M_m \cdots M_1 w_i\| \geq \exp(-\sum_j a_j S_{w_i}(T_j)/2)$. Averaging over i with weights p_i and applying concavity of the entropy gives $-2\log N \leq S(X) + \sum_j a_j S(T_j)$.

Subtracting the two bounds yields $\sum_j a_j\,[S(X_j) - S(T_j)] \leq S(X) + C$. Here $S(X_j) - S(T_j) = S(X|T_j)$, which is monotone decreasing in j by strong subadditivity. Since $\sum_j a_j = 2$, the discount follows. No property of the state on the flanks is used beyond their containing a constant number of sites, and no spectral gap of any restricted Hamiltonian is invoked.

Precisely. Fix $D = R$ (recall $R \geq 1$) and an integer $m \geq 4$. For an interval $X \subseteq \Lambda$ put

$$X_0 = X, \qquad X_j = X^{+jD}\ (1 \leq j \leq m),$$
$$T_j = X_j \setminus X, \qquad a_j = \frac{2}{m}.$$

Then $T_1 \subseteq \cdots \subseteq T_m$, each T_j consists of at most two intervals of length at most jD, and $\sum_j a_j = 2$. Define the constants

$$C_E := 8\,N_R\,J\,q^{2(R+1)},$$
$$m := \Big\lceil \max\Big\{4,\ \frac{4C_E}{\Delta},\ 128\sqrt{(1+\vartheta)\mathcal{B}_R},\ 4096\,\mathrm{e}\,(1+\vartheta)\,\mathcal{B}_R\Big\}\Big\rceil. \tag{14}$$

Every entry in the maximum is bounded by a function of q, R times $(1 + J/\Delta)$, so

$$m \leq C_1(q,R)\,(1+\vartheta). \tag{15}$$

Lemma IV.1 (Single splitting)**.** *A support Y with $\mathrm{diam}_{d_\Lambda} Y \leq R$ is split by at most one of the intervals $X_1, \ldots, X_m$. If it is split by X_j, it is contained in X_i for every $i > j$ and disjoint from X_i for every $i < j$. The same holds verbatim for the family $X_0, \ldots, X_m$.*

Proof. Suppose Y is split by X_j. Then Y contains a site $y \notin X_j$ and a site $x \in X_j$, and all sites of Y lie within graph distance $R \leq D$ of both x and y. Every site within distance D of X_j belongs to X_{j+1}, so $Y \subseteq X_{j+1} \subseteq X_i$ for all $i > j$. Conversely, every site of X_{j-1} lies at distance greater than D from $\Lambda \setminus X_j$ (since $X_j = X_{j-1}^{+D}$), so $Y \cap X_{j-1} = \varnothing$, and hence $Y \cap X_i = \varnothing$ for all $i < j$. A set that is contained in X_i or disjoint from X_i is not split by X_i. □

Lemma IV.2 (Buffered discount)**.** *With m as in (14), every interval $X \subseteq \Lambda$ satisfies*

$$S(X\,|\,T_m) \ \leq\ \frac{1}{2}\,S(X) + \frac{1}{2} + \frac{C_E}{\Delta}. \tag{16}$$

Proof. If $X = \varnothing$ both sides vanish. Assume $X \neq \varnothing$.

a. The optimisation. To ensure that the feasible sets are nonempty and every filter is invertible, fix $\epsilon > 0$ so small that $(\dim \mathcal{H}_{X_j})\epsilon^{2/a_j} < 1$ for all j, and maximise

$$N = \|L_m \cdots L_1 \Omega\|$$

over positive operators L_j on $\mathcal{H}_{X_j}$ (extended by the identity on $\Lambda \setminus X_j$) satisfying $L_j \geq \epsilon \mathrm{I}$ and $\mathrm{Tr}_{\mathcal{H}_{X_j}} L_j^{2/a_j} = 1$. The feasible sets are nonempty and compact and every feasible product is invertible, so a maximum $N > 0$ is attained. At a maximiser put $K = L_m \cdots L_1$, $\phi = K\Omega/N$, and $\rho_j = \mathrm{Tr}_{\Lambda \setminus X_j} |\phi\rangle\langle\phi|$.

b. Stationarity. To eliminate outer filters when evaluating expectations of operators supported in inner shells, we prove that $[L_j, \rho_j] = 0$ for all j by descending induction on j. The key mechanism is the stripping identity: if $[L_k, \rho_k] = 0$ and T is supported in X_k, then

$$\langle \phi, L_k T L_k^{-1} \phi\rangle = \mathrm{Tr}(L_k^{-1}\rho_k L_k T) = \mathrm{Tr}(\rho_k T) = \langle \phi, T\phi\rangle. \tag{17}$$

Assume $[L_k, \rho_k] = 0$ for all $k > j$ and let B be anti-Hermitian on $\mathcal{H}_{X_j}$. Replacing L_j by $\mathrm{e}^{tB} L_j \mathrm{e}^{-tB}$ preserves both constraints, so the derivative of $\log\|K\Omega\|$ at $t = 0$ vanishes. That derivative equals $\Re\langle \phi, K_{>j}(B - L_j B L_j^{-1}) K_{>j}^{-1}\phi\rangle$ with $K_{>j} = L_m \cdots L_{j+1}$. Because $B - L_j B L_j^{-1}$ is supported in $X_j \subseteq X_{j+1} \subseteq \cdots \subseteq X_m$, we can strip L_m by (17), then L_{m-1}, and so on down to L_{j+1}. Since $\mathrm{Tr}(\rho_j B)$ is purely imaginary, stationarity yields $\Re\,\mathrm{Tr}(L_j^{-1}\rho_j L_j B) = 0$ for all anti-Hermitian B. Hence $L_j^{-1}\rho_j L_j$ is Hermitian, i.e. equal to $L_j \rho_j L_j^{-1}$, so ρ_j commutes with L_j^2 and therefore with $L_j \geq 0$. No commutation between distinct filters is claimed or used.

To bound the eigenvalue ratios of L_j in terms of those of ρ_j, choose a common eigenbasis of L_j and ρ_j with eigenvalues l_i and p_i, and set $x_i = l_i^{2/a_j}$, so that $\sum_i x_i = 1$ and $x_i \geq \epsilon^{2/a_j}$. A diagonal variation of L_j together with the same stripping of $K_{>j}$ gives $d\log N = \frac{a_j}{2}\sum_i p_i\,dx_i/x_i$. By the Kuhn–Tucker conditions on the simplex with floor ϵ^{2/a_j}, the ratio p_i/x_i equals a constant $\lambda > 0$ on all coordinates above the floor and is at most λ on floored coordinates (at least one coordinate lies strictly above the floor by the choice of ϵ, and $\lambda > 0$ because shifting weight from a zero-probability free coordinate to a positive-probability floored coordinate would strictly increase N). Thus $x_i = \max\{\epsilon^{2/a_j}, p_i/\lambda\}$. Because taking a maximum with a common floor contracts differences of logarithms, we obtain

$$\Big|\log\frac{l_i}{l_k}\Big| \leq \frac{a_j}{2}\Big|\log\frac{p_i}{p_k}\Big| \qquad (p_i, p_k > 0). \tag{18}$$

c. Energy of the optimised vector. Since $KHK^{-1}\phi = E_0\phi$, the excitation energy of ϕ is

$$\langle \phi, (H - E_0)\phi\rangle = \sum_Y \Re\big(\langle \phi, h_Y \phi\rangle - \langle \phi, K h_Y K^{-1}\phi\rangle\big).$$

If Y is split by none of $X_1, \ldots, X_m$, then by Lemma IV.1 each X_j either contains Y or is disjoint from Y; stripping the containing filters from the outside in via (17)

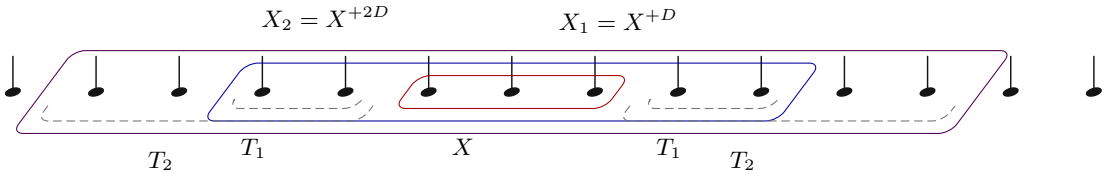


FIG. 4. The interval X and the nested shells $X_1 = X^{+D} \subset X_2 = X^{+2D}$, drawn with $D = R = 2$; the braces mark the flanks $T_j = X_j \setminus X$, each a union of at most two intervals of length at most jD, which carry the buffer.

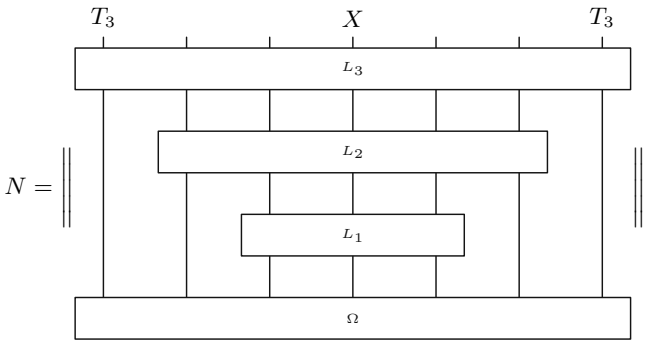


FIG. 5. The optimized norm. Three positive filters L_1, L_2, L_3 act on the nested intervals $X_1 \subset X_2 \subset X_3$ of the ground vector; each is normalized by $\operatorname{Tr} L_j^{2/a_j} = 1$.

and commuting the disjoint filters past h_Y shows that the summand vanishes. If Y is split by X_j, the same reduction strips $L_m, \dots, L_{j+1}$ and commutes $L_{j-1}, \dots, L_1$ past h_Y, leaving only the single-filter difference $\langle \phi, h_Y \phi \rangle - \langle \phi, L_j h_Y L_j^{-1} \phi \rangle$. Write $\phi = \sum_i \sqrt{p_i}\, e_i \otimes f_i$ for the Schmidt decomposition of ϕ across X_j, where the e_i are chosen in a common eigenbasis of ρ_j and L_j (diagonalising L_j within each eigenspace of ρ_j), and put $h_{ik} = \langle e_i \otimes f_i, h_Y (e_k \otimes f_k) \rangle$. Pairing (i,k) with (k,i) and using $h_{ki} = \overline{h_{ik}}$ gives

$$\begin{aligned} &\Re \langle \phi, L_j h_Y L_j^{-1} \phi \rangle - \langle \phi, h_Y \phi \rangle \\ &\qquad = \sum_{i,k:\, p_i p_k > 0} \sqrt{p_i p_k}\, \big(\cosh(\log(l_i/l_k)) - 1 \big) \Re h_{ik}. \end{aligned}$$

Set $z = \frac{1}{2} \log(p_i/p_k)$. By (18) and $a_j \le \frac{1}{2}$, we have $\cosh(\log(l_i/l_k)) - 1 \le \cosh(a_j z) - 1 \le \frac{1}{2} a_j^2 z^2 \cosh(a_j z) \le \frac{1}{2} a_j^2 z^2 \cosh(z/2)$, while $\sqrt{p_i p_k} = (p_i + p_k)/(2 \cosh z)$. Since $\sup_z z^2 \cosh(z/2)/(4 \cosh z) \le 1$, the prefactor of $\Re h_{ik}$ is at most $a_j^2 (p_i + p_k)$. Expanding h_Y in matrix units via (4) and applying (10) to each product term yields

$$\big| \Re \langle \phi, L_j h_Y L_j^{-1} \phi \rangle - \langle \phi, h_Y \phi \rangle \big| \le 2 a_j^2\, q^{2(R+1)} J.$$

Summing over $j = 1, \dots, m$ and the at most N_R interactions split by each X_j gives

$$\begin{aligned} 0 \le \langle \phi, (H - E_0) \phi \rangle &\le \sum_{j=1}^{m} N_R \cdot 2 a_j^2 q^{2(R+1)} J \\ &= \frac{8 N_R J q^{2(R+1)}}{m} = \frac{C_E}{m}. \end{aligned} \tag{19}$$

By the spectral gap (2) and the definition of m in (14),

$$1 - |\langle \Omega, \phi \rangle|^2 \le \frac{C_E}{m \Delta} \le \frac{1}{4}. \tag{20}$$

The exact same bound holds for the optimiser of every prefix $L_j \cdots L_1$ ($1 \le j \le m$): the stationarity and energy arguments apply verbatim to the j-shell problem with the same weights and floor, and any interaction split by X_k with $k > j$ is disjoint from $X_1, \dots, X_j$ and contributes zero, so the prefix excitation energy is bounded by the sum in (19) truncated to $i \le j$.

d. Upper bound on the optimum. To peel the filters off one by one, let N_j be the maximum norm for the prefix $1, \dots, j$ at the same floor ϵ (with $N_0 = 1$), and let $L_j \cdots L_1$ be a prefix-j maximiser with $K_{<j} = L_{j-1} \cdots L_1$. Applying (20) to the prefix-j maximiser and using Cauchy–Schwarz gives

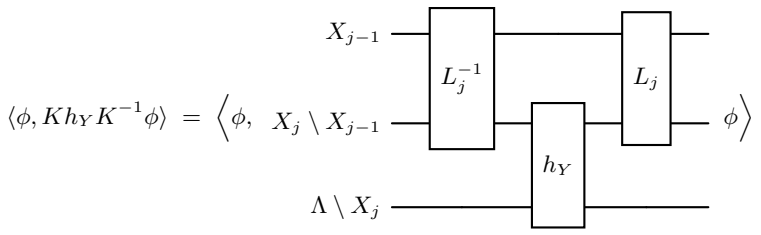
$$\begin{aligned} \sqrt{1 - C_E/(m\Delta)}\; N_j &\le |\langle \Omega, L_j K_{<j} \Omega \rangle| \\ &= |\langle L_j \Omega, K_{<j} \Omega \rangle| \le \|L_j \Omega\|\, N_{j-1}. \end{aligned}$$

For any feasible L_j, Schatten–Hölder with conjugate exponents $\frac{1}{1-a_j}$ and $\frac{1}{a_j}$ together with $\operatorname{Tr} L_j^{2/a_j} = 1$ gives

$$\|L_j \Omega\|^2 = \operatorname{Tr} \rho_{X_j} L_j^2 \le \big(\operatorname{Tr} \rho_{X_j}^{1/(1-a_j)} \big)^{1-a_j}.$$

Applying Lemma III.1 to the interval X_j at $u = -a_j/(1-a_j)$ (which satisfies $|u| \le 2a_j = 4/m \le u_0$ by (14)) and using $(1-a_j) u^2 \le 2 a_j^2$ yields

$$\log \|L_j \Omega\|^2 \le -a_j S(X_j) + 1024\, \mathrm{e}\, \vartheta\, \mathcal{B}_R\, a_j^2.$$



outer filters $L_{j+1}, \dots, L_m$: stripped by the stripping identity, $[L_k, \rho_k] = 0$
inner filters $L_1, \dots, L_{j-1}$: disjoint from Y, commute with h_Y

FIG. 6. Why the energy is small. An interaction h_Y split by X_j is conjugated by exactly one filter: the outer filters, whose regions contain Y, are stripped by the stripping identity at the optimum, and the inner ones commute with h_Y because they live on disjoint sites.

Multiplying the recursion $N_j \le (1 - C_E/(m\Delta))^{-1/2}\|L_j\Omega\|N_{j-1}$ over $j = 1, \dots, m$ and using $-\frac{1}{2}\log(1-v) \le v$ for $0 \le v \le \frac{1}{2}$, we obtain

$$\log N \le -\frac{1}{2}\sum_j a_j S(X_j) + 512\,\mathrm{e}\,\vartheta\,\mathcal{B}_R \sum_j a_j^2 + \frac{mC_E}{m\Delta}$$
$$\le -\frac{1}{2}\sum_j a_j S(X_j) + \frac{1}{2} + \frac{C_E}{\Delta},$$

since $\sum_j a_j^2 = 4/m \le 1/(1024\mathrm{e}\vartheta\mathcal{B}_R)$ by (14). Equivalently,

$$-2\log N \;\ge\; \sum_{j=1}^{m} a_j S(X_j) - 1 - \frac{2C_E}{\Delta}, \tag{21}$$

with constants independent of the floor ϵ.

e. Lower bound on the optimum by Schmidt pinning. Letting $\epsilon \downarrow 0$, the floored maxima converge to the supremum N over all nonnegative filters $L_j \ge 0$ satisfying $\mathrm{Tr}\, L_j^{2/a_j} = 1$ (since the feasible sets are compact and every nonnegative feasible filter is a limit of strictly positive feasible ones), and (21) continues to hold. Write $\Omega = \sum_i \sqrt{p_i}\, u_i \otimes w_i$ ($p_i > 0$) for the Schmidt decomposition across X. For each fixed i, test N on the trial filters

$$L_j = |u_i\rangle\langle u_i| \otimes M_j, \qquad M_j \ge 0 \text{ on } \mathcal{H}_{T_j},$$
$$\mathrm{Tr}\, M_j^{2/a_j} = 1,$$

which are feasible (since $\mathrm{Tr}_{X_j} L_j^{2/a_j} = \mathrm{Tr}\, M_j^{2/a_j} = 1$) and give $N \ge \sqrt{p_i}\, N_i$, where $N_i = \sup\|M_m \cdots M_1 w_i\|$ over such M_j. We claim that

$$\log N_i \ge -\frac{1}{2}\sum_j a_j S_{w_i}(T_j). \tag{22}$$

Because the flank filters $M_1, \dots, M_m$ act on nested regions $T_1 \subseteq \cdots \subseteq T_m$ and need not commute, we relate the product norm N_i to single-flank expectations by complex interpolation. Fix invertible feasible M_j and consider the analytic vector-valued function $F_i(z) = M_m^z \cdots M_1^z w_i$ on the strip $0 \le \Re z \le 1$. On $\Re z = 0$, the factors M_j^{iy} are unitary and $\|F_i(iy)\| = 1$. On $\Re z = 1$, set $U_j = M_j^{iy}$, $V_j = U_j \cdots U_1$, and $V_0 = \mathrm{I}$; collecting the unitaries to the left gives

$$M_m^{1+iy} \cdots M_1^{1+iy}$$
$$= (U_m \cdots U_1)\,(V_{m-1}^* M_m V_{m-1}) \cdots (V_0^* M_1 V_0).$$

Each conjugated factor $V_{j-1}^* M_j V_{j-1}$ is positive, supported in T_j (since V_{j-1} acts on $T_{j-1} \subseteq T_j$), and satisfies the same trace-power normalisation, so $\|F_i(1 + iy)\| \le N_i$. Testing $F_i(z)$ against arbitrary unit vectors and applying Hadamard's three-lines theorem yields $\log\|F_i(x)\| \le x \log N_i$ for $0 \le x \le 1$; differentiating at $x = 0^+$ gives $\sum_j \langle w_i, \log M_j\, w_i\rangle \le \log N_i$. Choosing $M_j = \big((1-\delta)\rho_{w_i,T_j} + \delta \mathrm{I}/\dim \mathcal{H}_{T_j}\big)^{a_j/2}$ and letting $\delta \downarrow 0$, the left-hand side converges to $-\frac{1}{2}\sum_j a_j S_{w_i}(T_j)$, proving (22). Consequently, $-2\log N \le -\log p_i + \sum_j a_j S_{w_i}(T_j)$ for every i. Averaging over i with weights p_i and using concavity of the entropy (since $\sum_i p_i \rho_{w_i,T_j} = \rho_{T_j}$) gives

$$-2\log N \le S(X) + \sum_{j=1}^{m} a_j S(T_j). \tag{23}$$

f. Conclusion. Subtracting (23) from (21) gives $\sum_j a_j\big(S(X_j) - S(T_j)\big) \le S(X) + 1 + 2C_E/\Delta$. Since $S(X_j) - S(T_j) = S(XT_j) - S(T_j) = S(X|T_j)$ and $S(X|T_j) \ge S(X|T_m)$ for all $j \le m$ by strong subadditivity (conditioning on the larger flank $T_m \supseteq T_j$ can only decrease conditional entropy), $\sum_j a_j = 2$ yields $2S(X|T_m) \le S(X) + 1 + 2C_E/\Delta$. □

V. CLOSING THE ARGUMENT

In words. Lemma IV.2 shows that $S(X|T) \le \frac{1}{2}S(X) + C$, where T is a buffer of a constant number of sites. On the other hand, conditional entropy cannot be arbitrarily negative: by the Araki–Lieb triangle inequality $S(XT) \ge S(X) - S(T)$, we have $S(X|T) \ge S(X) - 2S(T)$. Combining the two inequalities gives $S(X) - 2S(T) \le \frac{1}{2}S(X) + C$,

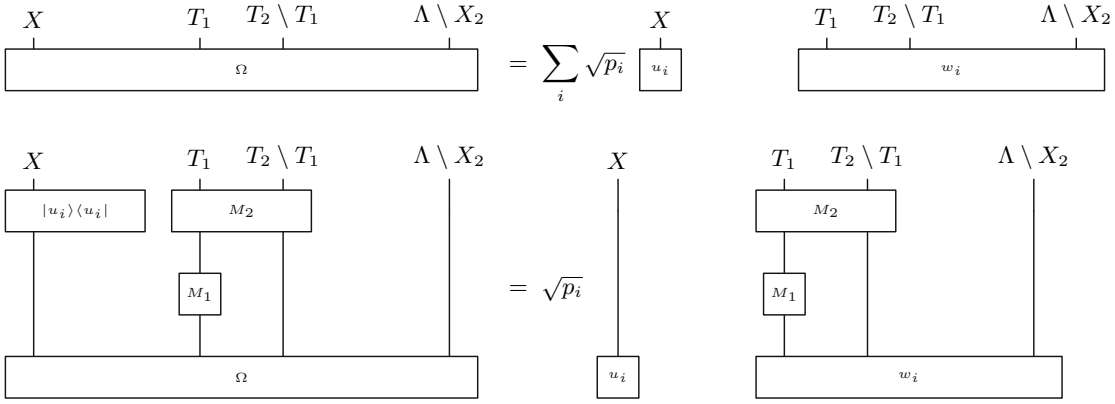


FIG. 7. Schmidt pinning. With the trial filters $|u_i\rangle\langle u_i| \otimes M_j$ (the projector is drawn once, by idempotence; the flanks are grouped into one leg each), the filtered vector becomes $\sqrt{p_i}\,u_i$ tensored with the flank filters applied to the partner w_i, which lower-bounds the optimum by $-S(X) - \sum_j a_j S(T_j)$ in the exponent after averaging over i.

hence $S(X) \le 4S(T) + 2C$. Since $S(T)$ is bounded by the number of sites of T times $\log q$, the area law follows immediately.

Nothing further is needed on a line. In two dimensions, the same pair of inequalities for a square X of side r and a buffer T of width proportional to r gives $S(X) \le 4S(T)+Cr$; there T is an annulus of area $\sim r^2$ whose entropy is as unknown as that of X, so the estimate does not close. That geometric obstruction is the exact reason the two-dimensional proof requires the multi-scale machinery of [1].

Precisely.

Proof of Theorem I.1. By Lemma II.1 it suffices to bound $S(X)$ for an interval X in an interval domain. Let $T = T_m$ be the buffer of Lemma IV.2, which contains at most $2mD$ sites. The Araki–Lieb inequality $S(XT) \ge S(X) - S(T)$ gives $S(X|T) \ge S(X) - 2S(T)$, and (16) gives $S(X|T) \le \frac{1}{2}S(X) + \frac{1}{2} + C_E/\Delta$. Hence

$$S(X) \le 4S(T)+1+\frac{2C_E}{\Delta} \le 8mD\log q+1+\frac{2C_E}{\Delta} =: C_{\mathrm{int}}. \tag{24}$$

By (15) and (14), $C_{\mathrm{int}} \le C_0(q,R)(1 + J/\Delta)$ with, for instance, $C_0(q,R) = 8C_1(q,R)\,R\log q+1+16N_Rq^{2(R+1)}$. □

Remark V.1 (What the constant depends on)**.** The spectral gap enters only through $\vartheta = J/\Delta$, linearly. The dependence on the local dimension and interaction range enters through $N_R\,q^{2(R+1)}$, which for nearest-neighbour interactions ($R = 1$) is of order q^4. The constants in Section IV were chosen for simplicity rather than sharpness; Remark V.2 shows that the polynomial dependence on q is an artifact of bounding the matrix-unit expansion term by term. By comparison, the bound of Arad–Kitaev–Landau–Vazirani [4] is $\tilde{O}(\log^3(q)/\Delta)$ for $R = 1$.

Remark V.2 (Sharper constants, and what is intrinsic)**.** Three aspects of the constants in (24) are not intrinsic to the argument.

The count N_R. The bound $2 \cdot 2^{2R}$ allows a separate interaction term for every subset of a $2R$-site window. When the Hamiltonian has one term per window of $R+1$ consecutive sites, a cut edge splits at most R of them, so N_R may be replaced by $2R$.

The factor $q^{2(R+1)}$ in C_E. This factor enters (19) because the matrix-unit expansion (4) was bounded term by term. In the proof of Lemma III.1, the same dimensional factor was converted into a logarithm via Hadamard's three-lines theorem, and the identical device applies here. For a split term $h = h_Y$ and the optimal filter $L = L_j$, define the entire function $g(z) = \langle\phi, L^z h L^{-z}\phi\rangle$. Then $g(0) = \langle\phi, h\phi\rangle$, $g(1) = \langle\phi, LhL^{-1}\phi\rangle$, $|g(iy)| \le J$ because L^{iy} is unitary, and $g(-\bar{z}) = \overline{g(z)}$. In the common eigenbasis of ρ_j and L_j, $g(z) = \sum_{i,k}\sqrt{p_ip_k}\,(l_i/l_k)^z h_{ik}$, and (18) together with the weighted arithmetic–geometric mean inequality gives $\sqrt{p_ip_k}\,|(l_i/l_k)^z| \le p_i + p_k$ for $|\Re z| \le 1/a_j$; hence $|g(z)| \le 2Jq^{2(R+1)}$ on that strip by (10). Since $\log(2q^{2(R+1)}) \le 2\ell_R$, the three-lines theorem gives $|g(z)| \le \mathrm{e}J$ for $|\Re z| \le 1/(2a_j\ell_R)$, and Cauchy's estimate as in (12) yields, because $a_j \le 1/(4\ell_R)$,

$$|\Re\big(g(1) - g(0)\big)| \le 8\,\mathrm{e}\,J\,\ell_R^2\,a_j^2.$$

Thus (19) holds with $2a_j^2q^{2(R+1)}J$ replaced by $8\mathrm{e}J\ell_R^2a_j^2$, i.e. with

$$C'_E = 32\,\mathrm{e}\,N_R\,J\,\ell_R^2 \le 32\,\mathrm{e}\,J\,\mathcal{B}_R$$

in place of C_E. The entry $4C'_E/\Delta$ in (14) is then dominated by $4096\mathrm{e}(1+\vartheta)\mathcal{B}_R$, and (24) becomes

$$S(X) \;\le\; C\,(1 + J/\Delta)\,\mathcal{B}_R\,R\log q,$$

where now $\mathcal{B}_R \le 1 + 2R\big(1 + (R+1)\log q\big)^2$ and C is an absolute constant. For nearest-neighbour chains this is $O\big((1 + J/\Delta)\log^3 q\big)$, matching the scaling of the Arad–Kitaev–Landau–Vazirani bound [4] without using a Lieb–Robinson bound or an approximate ground-state projector. For general range R it gives $O\big((1+J/\Delta)R\log^3(q^R)\big)$,

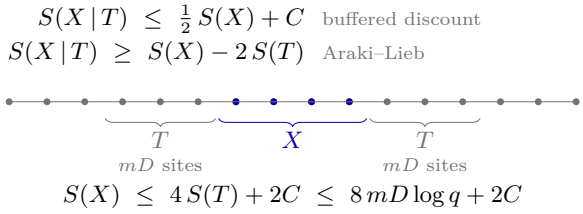


FIG. 8. The closing step. The interval X with its constant-width buffers T; the discount bounds $S(X|T)$ from above, ArakiLieb from below, and T has at most $2mD$ sites.

one factor of R above the blocked nearest-neighbour bound.

What is intrinsic. Two features of the constants cannot be removed by sharper estimates within this variational framework. First, the linear dependence on J/Δ: the Schmidt-pinning bound (23) carries $S(X)$ with coefficient one, so closing the inequality requires a total filter weight $\sum_j a_j > 1$. Spreading a total weight of order one across m shells costs the filtered vector an excitation energy of order $J\mathcal{B}_R/m$ by (19); the overlap bound (20) requires that energy to lie below the spectral gap Δ, forcing $m \gtrsim J\mathcal{B}_R/\Delta$, so the buffer entropy $S(T) \le 2mR\log q$ scales linearly with J/Δ. Second, the width $u_0 \sim (\vartheta\mathcal{B}_R)^{-1/2}$ of the tail window in Lemma III.1: the tilted vector of (13) has excitation energy $O(J\mathcal{B}_R u^2)$ for all $|u| \le 1/(16\ell_R)$, but the gap is used only through the single-shot variational inequality (2), which becomes vacuous once the excitation energy exceeds Δ. That window width sets the exponential decay rate of the Schmidt tail (8) and hence the exponent of L in the MPS bond dimension of Theorem VII.2 (Section VII), which is of order $\sqrt{(1+J/\Delta)\mathcal{B}_R}$. In the approximate-ground-state-projector construction of [4], the gap-dependent quantities sit in a prefactor while the exponent of the chain length is universal; moving $\sqrt{J/\Delta}$ out of the exponent would require applying the spectral gap in an amplified polynomial form, which is precisely what an approximate ground-state projector provides.

Remark V.3 (What was used)**.** The proof of Theorem I.1 uses the global spectral gap in only two places: in Lemma III.1 applied at the cuts of the shells X_j, and to convert the excitation energy bound (19) into the ground-state overlap (20). No Lieb–Robinson bound, no spectral filter, no approximate ground-state projector, no Chebyshev polynomial, and no bootstrapping of a Schmidt-rank bound appear. The von Neumann entropy itself enters only through operator norms of filtered vectors.

VI. THE LONG ROUTE, FOR COMPARISON

In words. It is instructive to trace how the multi-scale architecture of [1] would operate on a line if one refrained from closing the argument directly with the Araki–Lieb inequality, both to see where each two-dimensional ingredient enters and because the amplification and two-family steps reappear in spirit in Part II.

Sub-volume exponent. Partition an interval of length Mr into M subintervals ("children") of length r and reveal them in a uniformly random permutation using the chain rule for entropy. Whenever all children intersecting the constant-width buffer T_i of an interior child X_i have already been revealedan event of fixed positive probability $p_0$the conditional entropy of X_i is discounted to $S(X_i|T_i) \le \frac{1}{2}S(X_i)+C$. Averaging over the permutation yields the recursion $F(Mr) \le (1-p_0/4)M\,F(r)+CM$ for the worst-case interval entropy $F(r)$, which iterates to $F(r) \le Cr^{e_0}$ with a sub-volume exponent $e_0 < 1$. In two dimensions the analogous recursion gives r^{1+e_0}, which still exceeds the boundary r; in one dimension r^{e_0} already lies strictly below the interval length r, which is the threshold required for amplification.

Amplification. The amplification mechanism of [1, Section 10] converts a sub-linear mutual-information bound $I(X : \text{far}) \le n^{1-\varepsilon}$ into rapid polynomial decay $I(X : J) \le n^{-M}$ for any region J at graph distance at least $2r$ from X, where $r = O((n^{1-\varepsilon}+\log n)^{1/\alpha})$ with $\alpha < 1$ chosen as close to 1 as desired, using positive quasi-local constraints built from Kitaev's spectral filter, a random product of the associated contractions, and a polar decomposition. On a line, the elementary bound $I(X : \text{anything}) \le 2S(X) \le Cn^{e_0}$ for an interval X of length n directly provides the required sub-linear input without any coarse collar: choosing $\alpha > e_0$ yields $I(X : J) \le n^{-M}$ beyond distance $Cn^{e_0/\alpha} = o(n)$.

Two families. To bound the entropy of a half-chain A separated by a cut from its complement B, remove a constant-width block D adjacent to the cut and partition $A \setminus D$ into dyadic intervals $X_i = [2^i, 2^{i+1})$ with alternating family labels (odd i in one family, even i in the other). Each interval X_i is separated from B and from all earlier intervals of its own family by the preceding interval X_{i-1}, whose length 2^{i-1} far exceeds the amplification radius $O(2^{ie_0/\alpha})$. The two-family subadditivity cancellation of [1] then gives $S(A) \le |D|\log q + \frac{1}{2}\sum_i 2^{-iM}$ with $|D| = O(1)$.

Thus the multi-scale route also closes on a line, with the entire replica scanner replaced by the trivial inequality $I \le 2S$; the direct Araki–Lieb closure of Section V is simply much shorter and yields sharper constants.

Precisely. We record only the sub-volume recursion, as the amplification and two-family steps are direct applications of the propositions of [1] with the inputs above. Let $F(r)$ be the supremum of $S(X)$ over all intervals $X \subseteq \Lambda$ of length at most r (across all admissible Hamiltonians satisfying (2)), and let $w_0 = mD$ be the buffer width of

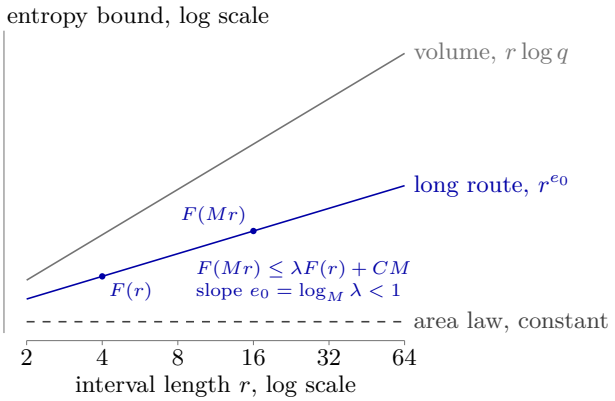


FIG. 9. Entropy bounds for an interval of length r on loglog axes (drawn with $M = 4$): the volume bound $r \log q$, parallel to the diameter r, has slope one; the long route gives slope $e_0 < 1$; the area law is the constant line reached directly in the previous stage.

Lemma IV.2 with discount constant $C_{\mathrm{buf}} = \frac{1}{2} + C_E/\Delta$. Fix $M \ge 8(2w_0/r + 2)$ and let X be an interval of length at most Mr. Partition X into at most M consecutive subintervals (children) of length at most r and order them by a uniformly random permutation. Call a child *interior* if its buffer of width w_0 on each side lies entirely within X; at most $B := 2\lceil w_0/r \rceil + 2$ children fail to be interior. For each interior child X_i, the event that all children intersecting its buffer T_i precede X_i in the permutation has probability at least $p_0 = (2\lceil w_0/r \rceil + 3)^{-1}$; on that event, strong subadditivity and Lemma IV.2 bound its chain-rule contribution by $S(X_i|T_i) \le \frac{1}{2} S(X_i) + C_{\mathrm{buf}}$, while on the complementary event it is at most $S(X_i)$ (since the event's probability is only bounded from below, the additive constant C_{buf} is charged in full). Taking the expectation of the entropy chain rule over the random permutation gives

$$\begin{aligned} F(Mr) &\le \Big[(1 - \tfrac{p_0}{2})M + \tfrac{p_0}{2} B\Big] F(r) + M C_{\mathrm{buf}} \\ &\le \big(1 - \tfrac{p_0}{4}\big) M\, F(r) + CM, \end{aligned}$$

using $B \le M/2$. Setting $\lambda = (1 - p_0/4)M \in (1, M)$ for $r \ge w_0$ and iterating along the geometric scales $r = w_0, Mw_0, M^2 w_0, \dots$ gives $F(M^j w_0) \le C\lambda^j$, and hence $F(r) \le C r^{e_0}$ with $e_0 = \log_M \lambda < 1$.

Part II
The one-dimensional matrix product state approximation

VII. STATEMENT, AND THE TWO-LINE ROUTE

In words. Theorem I.2 asks for a single global matrix product state approximating the ground vector, rather than an entropy bound on each cut individually. An entropy bound alone does not yield a rank truncation because von Neumann entropy controls only the mean of the entanglement spectrum, not the weight of the small Schmidt eigenvalues discarded when truncating a cut; in two dimensions, moreover, truncations across different cuts do not assemble into a consistent tensor network, and a distributed protocol is needed to bridge the gap. In one dimension there is a classical shortcut: if the Schmidt spectrum across every prefix cut has a rapidly decaying tail, then truncating the cuts sequentially from one end of the chain to the other produces a matrix product state whose global vector error is bounded by the sum of the square roots of the discarded tail weights (Verstraete–Cirac). The marginal tail bound of Section III provides precisely such an exponential tail at a rate u_0 depending only on $q, R, J/\Delta$, while the area law fixes the mean of the spectrum. Combining the two yields Theorem I.2 in a few lines, with bond dimension CL^{4/u_0}.

We record this sequential-truncation proof first because it is direct and elementary. The remainder of Part II then develops the one-dimensional *protocol* proof, which avoids sequential cut truncations altogether and is the construction that generalises to two dimensions.

Precisely.

Proposition VII.1 (Sequential truncation)**.** *Let Ω be a unit vector on $\mathcal{H}_\Lambda$, $\Lambda = \{1, \dots, L\}$. For $1 \le k \le L - 1$ let Π_k be the projection of $\mathcal{H}_{[1,k]}$ onto the span of the $\min\{D, \dim \mathcal{H}_{[1,k]}\}$ eigenvectors of $\rho_{[1,k]}$ with largest eigenvalues (so $\Pi_k = \mathrm{I}$ when $D \ge \dim \mathcal{H}_{[1,k]}$), let $\epsilon_k = \langle \Omega, (\mathrm{I} - \Pi_k \otimes \mathrm{I})\Omega \rangle$ be the discarded mass, and put*

$$\Phi = (\Pi_1 \otimes \mathrm{I})(\Pi_2 \otimes \mathrm{I}) \cdots (\Pi_{L-1} \otimes \mathrm{I})\, \Omega.$$

Then Φ is a matrix product state with open boundary conditions and bond dimension at most D, and $\|\Phi - \Omega\| \le \sum_k \sqrt{\epsilon_k}$.

Proof. Write $P_k = \Pi_k \otimes \mathrm{I}$. Telescoping gives $\Phi - \Omega = \sum_{k=1}^{L-1} P_1 \cdots P_{k-1}(P_k - \mathrm{I})\Omega$, and since each prefix $P_1 \cdots P_{k-1}$ is a contraction, the triangle inequality gives $\|\Phi - \Omega\| \le \sum_k \|(P_k - \mathrm{I})\Omega\| = \sum_k \sqrt{\epsilon_k}$. To bound the bond dimension, fix a cut k. The intermediate vector $P_k P_{k+1} \cdots P_{L-1}\Omega$ lies in $\operatorname{ran} \Pi_k \otimes \mathcal{H}_{[k+1,L]}$ and therefore has Schmidt rank at most D across cut k; the remaining factors $P_1, \dots, P_{k-1}$ act solely on $[1, k-1]$ to the left of cut k and cannot increase that Schmidt rank. A vector with Schmidt rank at most D across every cut is a matrix product state of bond dimension at most D. □

Theorem VII.2 (Theorem I.2, two-line route)**.** *Under the hypotheses of Theorem I.2, for every $L \ge 2$ there is a matrix product state Φ with bond dimension at most $2\,\mathrm{e}^{C_{\mathrm{int}}}\,(32\mathrm{e}^{\mathrm{e}/2} L^4)^{1/u_0}$ and $\min_\theta \|\Phi/\|\Phi\| - \mathrm{e}^{i\theta}\Omega\| \le L^{-1}$, where u_0 is (6) and C_{int} is (24).*

Proof. Fix a cut k and put $X = [1, k]$. By Lemma III.1, the eigenvalues p of ρ_X with $-\log p > S(X) + w$ have total mass at most $2\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w}$, while the eigenvalues with $-\log p \le S(X) + w$ each satisfy $p \ge \mathrm{e}^{-S(X)-w}$

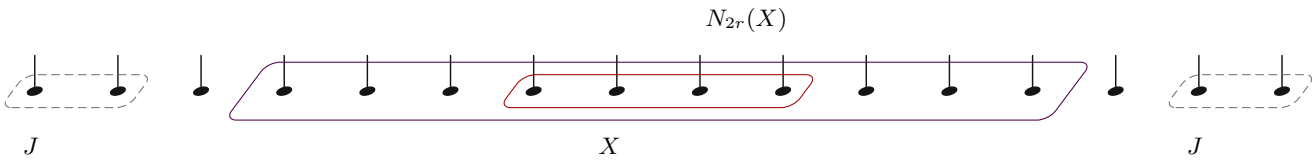


FIG. 10. Amplification on the chain: the interval X of length n, its graph neighbourhood $N_{2r}(X)$ with $r = O(n^{e_0/\alpha}) = o(n)$, on which the local unitary of the amplification argument acts, and the far set J it never touches, so that $I(X : J) \le n^{-M}$.

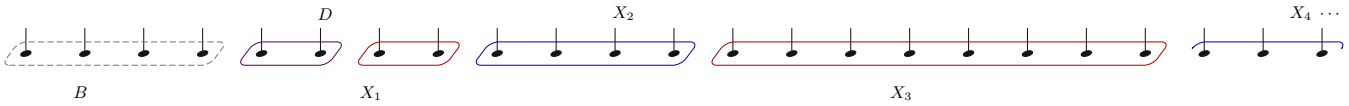


FIG. 11. Two families on a half-line: beyond the cut and the removed sites D, the dyadic intervals $X_i = [2^i, 2^{i+1})$ of lengths $2, 4, 8, \ldots$ with alternating family labels, continuing to the end of the chain. Each is separated from B and from all earlier same-family intervals by the previous interval, of half its length.

and therefore number at most $\mathrm{e}^{S(X)+w}$. Choose w so that $2\mathrm{e}^{\mathrm{e}/2}\mathrm{e}^{-u_0 w} = (16L^4)^{-1}$ and set $D = \lceil \mathrm{e}^{C_{\mathrm{int}}+w} \rceil \ge \mathrm{e}^{S(X)+w}$ by Theorem I.1. Then $\epsilon_k \le (16L^4)^{-1}$ for every k, and Proposition VII.1 gives $\|\Phi - \Omega\| \le (L-1)/(4L^2) \le 1/(4L)$. In particular $\Phi \ne 0$ and $\|\Phi/\|\Phi\| - \Omega\| \le 2\|\Phi - \Omega\| \le L^{-1}$. The bond dimension satisfies $D \le 2\mathrm{e}^{C_{\mathrm{int}}+w}$ with $\mathrm{e}^w = (32\mathrm{e}^{\mathrm{e}/2}L^4)^{1/u_0}$. □

Remark VII.3. The exponent $4/u_0$ is of order $\sqrt{(1 + J/\Delta)\mathcal{B}_R}$. Whereas Hastings' original construction yields a polynomial bond dimension whose exponent is exponential in J/Δ, the marginal tail bound gives a square-root dependence on J/Δ. This exponent is a feature of the direct sequential-truncation route rather than of the protocol below.

Plan of the protocol route

In words. The distributed protocol proof has four layers:

1. *Information at logarithmic separation.* Two regions of the chain separated by $D \log L$ sites share at most L^{-p} nats of mutual information. This is where the protocol route uses the spectral gap. In two dimensions, establishing logarithmic separation requires a coarse collar bound, a cell conditional-information lemma, and an angular induction; in one dimension, it follows directly from the area law and a single two-replica information-reset lemma (Section VIII).
2. *Ownership moves.* Three elementary operations redistribute site registers among parties while keeping the global state close to a reference state: a *birth* (handing an interior interval to a new owner), its reverse, a *death*, and an *exchange* of ownership assignments between two copies of the ground state inside a window. Each move is a contraction built from private maps and at most one normalised shared bipartite vector or covector, and each incurs $O(L^{-p/2})$ vector error whenever the moved region has mutual information at most L^{-p} with all third-party registers (Section IX).
3. *The sweep schedule.* A left-to-right sweep with a moving remainder party creates the logarithmic block owners one by one. The only obstruction is dissolving a narrow remainder band wedged between two distinct neighbouring owners; this is accomplished by a fixed sequence of moves using an auxiliary copy of the ground state, copied from the edge construction of the two-dimensional paper [2], so that every party participates in a bounded number of moves (Section X).
4. *Compression and assembly.* The sweep is a linear calculation whose intermediate private registers have arbitrarily large dimension. Replacing each shared bipartite resource in the density calculation by an empirical average of polynomially many random product operators approximates the output physical density in trace norm; the error bound is independent of the private dimensions because each party participates in boundedly many gates. Extracting a single computational-basis column of the resulting operator network and placing its nodes along the chain yields the matrix product state (Sections XI, XII).

VIII. INFORMATION AT LOGARITHMIC SEPARATION

A. Purifications, fidelity, and splitting

In words. To change the ownership of a region T without disturbing the global state, we must decouple T from the registers E held by third parties using an operation supported entirely on the separating buffer $U = \Lambda \setminus (T \cup E)$. Small mutual information $I(T : E)$ means that the reduced state ρ_{TE} is close in relative entropy to the product state $\rho_T \otimes \rho_E$. Relative entropy controls Uhlmann fidelity, which equals the maximum overlap between purifications. Since U purifies TE in the pure ground state Ω, Uhlmann's theorem guarantees an isometry V acting on the buffer U alone that maps Ω into nearly a tensor product $s \otimes s'$ of a purification of ρ_T and a purification of ρ_E. Any party holding U

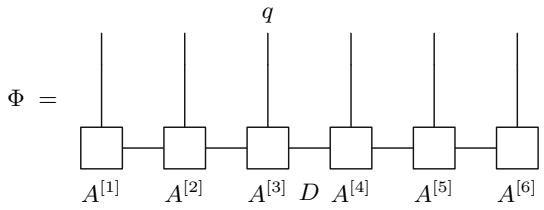


FIG. 12. A matrix product state with open boundary conditions: one tensor per site, bonds of dimension D between neighbours, one physical leg per site.

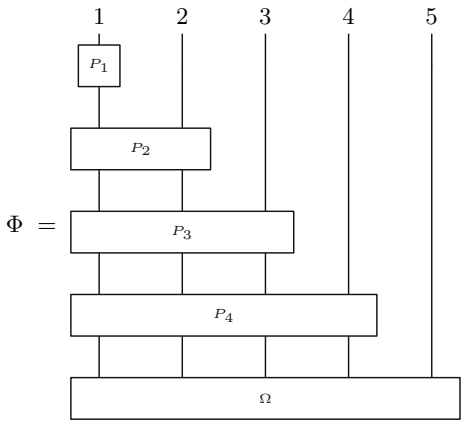


FIG. 13. Sequential truncation, drawn for $L = 5$. The projector $P_k = \Pi_k \otimes 1$ onto the top D Schmidt vectors of the prefix $[1, k]$ is applied for $k = L - 1$ first and $k = 1$ last; the output of P_k lies in $\operatorname{ran}\Pi_k \otimes \mathcal{H}_{[k+1,L]}$, so across every cut the result has Schmidt rank at most D.

can execute this decoupling privately; this single mechanism underlies all births, deaths, and exchanges.

Precisely. For density operators ρ, σ put $F(\rho, \sigma) = \|\sqrt{\rho}\sqrt{\sigma}\|_1$ and $D(\rho\|\sigma) = \operatorname{Tr}\rho(\log\rho - \log\sigma)$, taken to be infinite when the support of ρ is not contained in that of σ.

Lemma VIII.1 (Fidelity, information, Uhlmann)**.** *For normalised finite-dimensional states, $F(\rho, \sigma) \geq \mathrm{e}^{-D(\rho\|\sigma)/2}$, and $D(\rho_{AC}\|\rho_A \otimes \rho_C) = I(A : C)$. If ξ purifies ρ on $\mathcal{H} \otimes \mathcal{K}$ and η purifies σ on $\mathcal{H} \otimes \mathcal{K}'$, then after enlarging $\mathcal{K}'$ if necessary there is an isometry $V : \mathcal{K} \to \mathcal{K}'$ with $|\langle\eta, (\mathrm{I} \otimes V)\xi\rangle| = F(\rho, \sigma)$.*

Proof. Let $(p_i, |i\rangle)$ and $(q_j, |j'\rangle)$ be eigenpairs of ρ and σ. If $D(\rho\|\sigma) < \infty$, then $w_{ij} = p_i|\langle i|j'\rangle|^2$ defines a probability distribution on the pairs with $p_i q_j > 0$, and $D(\rho\|\sigma) = \sum_{ij} w_{ij} \log(p_i/q_j)$. By convexity of the exponential, $\operatorname{Tr}\sqrt{\rho}\sqrt{\sigma} = \sum_{ij} w_{ij} \exp(-\frac{1}{2}\log(p_i/q_j)) \geq \mathrm{e}^{-D(\rho\|\sigma)/2}$, and the trace norm dominates the trace. The identity $D(\rho_{AC}\|\rho_A \otimes \rho_C) = I(A : C)$ follows from expanding $\log(\rho_A \otimes \rho_C)$. The purification statement is Uhlmann's theorem: the overlaps $\operatorname{Tr}(\sqrt{\rho}\sqrt{\sigma}U)$ over unitaries U on a common purifying space have maximum modulus $\|\sqrt{\rho}\sqrt{\sigma}\|_1$ by polar decomposition, and any two purifications of the same state are related by an isometry on the purifying factor. □

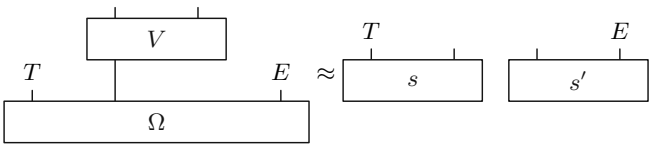


FIG. 14. The splitting lemma as a diagram: the isometry V on the buffer U turns Ω into nearly a product of a purification s of ρ_T and a purification s' of ρ_E.

Lemma VIII.2 (Splitting)**.** *Let T, U, E partition Λ and $b = I_\Omega(T : E)$. There are finite-dimensional spaces $\mathcal{B}_T, \mathcal{B}_E$, an isometry $V : \mathcal{H}_U \to \mathcal{B}_T \otimes \mathcal{B}_E$ and unit vectors $s \in \mathcal{H}_T \otimes \mathcal{B}_T$, $s' \in \mathcal{H}_E \otimes \mathcal{B}_E$ with*

$$\|(\mathrm{I}_{TE} \otimes V)\Omega - s \otimes s'\| \leq \sqrt{2(1 - \mathrm{e}^{-b/2})} \leq \sqrt{b}.$$

Proof. Because ρ_{TE} is supported in $\operatorname{supp}\rho_T \otimes \operatorname{supp}\rho_E$ (the projector onto the kernel of a marginal, tensored with the identity, has zero expectation in the positive joint state), the relative entropy $D(\rho_{TE}\|\rho_T \otimes \rho_E) = b$ is finite. Choose purifications s of ρ_T and s' of ρ_E, enlarge their purifying spaces so that $\mathcal{B}_T \otimes \mathcal{B}_E$ contains an isometric copy of $\mathcal{H}_U$, and apply Lemma VIII.1: there is an isometry V on $\mathcal{H}_U$ such that $(\mathrm{I}_{TE} \otimes V)\Omega$ has real overlap at least $\mathrm{e}^{-b/2}$ with $s \otimes s'$. Two unit vectors with real overlap F have Euclidean distance $\sqrt{2(1 - F)}$. □

B. The information reset

In words. The area law gives $I(R:C)=O(1)$ between an interval R and any disjoint region C, never $o(1)$. To assemble a global tensor network from local operations across the chain, however, errors must be suppressed to an arbitrary inverse power L^{-p} so that they remain small when summed over $O(L)$ steps. The information-reset lemma bridges this gap by shrinking R and C inward to cores $R_0 \subseteq R$ and $C_0 \subseteq C$ separated from their outer boundaries by a margin of width $w = O(\log L)$.

The physical mechanism is a two-replica swap. Take two identical copies of the ground vector, $\Psi = \Omega \otimes \Omega$, and swap the two copies on R to obtain $\Phi = F_R \Psi$. In the first replica of Φ, the core $R_0 \subseteq R$ has been swapped in from the second copy of Ω while $C_0 \subseteq C$ remains from the first, so the two cores originate from independent copies and have the exact product marginal $\rho_{R_0} \otimes \rho_{C_0}$. If a unitary U supported entirely in the buffer region between R_0 and C_0 maps Ψ to Φ up to error $O(L^{-r_*})$, then the joint marginal of the cores in Ψ agrees with $\rho_{R_0} \otimes \rho_{C_0}$ up to $O(L^{-r_*})$, which implies $I_\Omega(R_0 : C_0) \le L^{-p}$.

The unitary U is constructed in three steps. First, the bound $I(R:C) \le b$ and the splitting lemma produce a Hermitian contraction W on the buffer $Q = \Lambda \setminus (R \cup C)$ of both copies with macroscopic overlap $z = \langle \Phi, W\Psi \rangle \ge \mathrm{e}^{-2b}$. Because $W\Psi$ still carries excited components, W is not yet close to a unitary mapping Ψ to Φ. Second, a two-sided Gaussian spectral filter, $M = \int g_h(t) \mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}} dt$, damps out excited states in both directions, projecting $W\Psi$ onto $z\Phi$ and $W^\dagger \Phi$ onto $z\Psi$ with error $\mathrm{e}^{-h\Delta^2/2}$, because Ψ and Φ are the unique gapped ground vectors of the doubled Hamiltonians $\mathbb{H}$ and $\mathbb{H}' = F_R \mathbb{H} F_R$. Third, because time evolution spreads operator supports, a Lieb–Robinson bound localises M to within distance $\ell = O(1 + b + \log L)$ of the buffer Q and of the boundary zone Z_R where $\mathbb{H}$ and $\mathbb{H}'$ differ. Rescaling the localised operator makes it almost isometric on Ψ, and its polar unitary U remains supported strictly outside both cores. The lemma is proved exactly as in the two-dimensional source; we only insert the one-dimensional locality bound.

Precisely. To control the spatial spread of the Gaussian filter, we first record the standard Lieb–Robinson propagation estimate on the doubled chain. Let $\mathbb{H}$ be a Hamiltonian on $\bigotimes_{x\in\Lambda}(\mathbb{C}^q \otimes \mathbb{C}^q)$ written as a sum of terms k_Z supported on sets Z of diameter at most R with $\sum_{Z \ni x} \|k_Z\| \le b_0$ for every x, and let $\tau_t(A) = \mathrm{e}^{it\mathbb{H}} A \mathrm{e}^{-it\mathbb{H}}$.

Lemma VIII.3 (Lieb–Robinson bound on a chain)**.** *There are $C, v, \mu > 0$ depending only on R and b_0 such that for every operator A supported in $S \subseteq \Lambda$, every site y, and every operator B_y supported at y,*

$$\|[\tau_t(A), B_y]\| \le C|S|\,\|A\|\,\|B_y\|\,\mathrm{e}^{v|t| - \mu\,\mathrm{dist}(S,y)}.$$

Consequently, if $\mathcal{P}_D$ denotes the conditional expectation onto operators supported in D (the average over independent Haar unitaries at the sites outside D), then for every $\ell \ge 0$,

$$\|\tau_t(A) - \mathcal{P}_{N_\ell(S)} \tau_t(A)\| \le C'|S|^2\,\|A\|\,\mathrm{e}^{v|t| - \mu\ell}.$$

Proof. For $t \ge 0$ put $c_y(S,t) = \sup\{\|[\tau_t(A), B_y]\| : \operatorname{supp} A \subseteq S, \|A\| \le 1\}$. Differentiating and using the Jacobi identity shows that $\frac{d}{dt}[\tau_t(A), B_y]$ equals $i[\sum_{Z\cap S\neq\varnothing} \tau_t(k_Z), [\tau_t(A), B_y]]$ plus a sum over $Z \cap S \neq \varnothing$ of $-i[\tau_t(A), [\tau_t(k_Z), B_y]]$. The first term is removed by a unitary change of frame, yielding the integral inequality $c_y(S,t) \le c_y(S,0) + 2\sum_{Z\cap S\neq\varnothing} \|k_Z\| \int_0^t c_y(Z,u)\,du$. Iterating this inequality, and using the trivial bound $c_y(Z,u) \le 2\|B_y\|$ to see that the remainder after m iterations vanishes as $m \to \infty$, gives

$$c_y(S,t) \le 2\|B_y\| \sum_{m\ge 0} \frac{(2t)^m}{m!} \sum_{\substack{Z_1\cap S\neq\varnothing,\ Z_{i+1}\cap Z_i\neq\varnothing \\ y\in Z_m}} \prod_{i=1}^{m} \|k_{Z_i}\|,$$

with the $m = 0$ term equal to $\mathbf{1}_{\{y\in S\}}$. Since the norms of the interaction terms meeting any set of n sites sum to at most nb_0, the inner sum is at most $|S|\,b_0\,((R+1)b_0)^{m-1} \le |S|\kappa^m$ with $\kappa = (R+1)b_0$. Moreover, a chain of m supports of diameter at most R connecting S to y requires $mR \ge \mathrm{dist}(S,y)$. Hence, for every $\mu > 0$, $c_y(S,t) \le 2|S|\|B_y\| \mathrm{e}^{-\mu\,\mathrm{dist}(S,y)} \sum_m (2\kappa \mathrm{e}^{\mu R} t)^m / m!$, which gives the first bound with $v = 2\kappa \mathrm{e}^{\mu R}$; negative times are identical. For the second bound, expressing $\mathcal{P}_D$ as a product of commuting single-site Haar twirls T_y and telescoping gives $\|B - \mathcal{P}_D B\| \le \sum_{y\notin D} \sup_{U_y} \|[B, U_y]\|$. Because at most $2|S|$ sites lie at each distance $d > \ell$ from S, the sum is bounded by $4C|S|^2\|A\|\mathrm{e}^{vt}\sum_{d>\ell} \mathrm{e}^{-\mu d}$. □

Lemma VIII.4 (Information reset)**.** *For every $p > 0$ there is $C_p < \infty$, depending only on p, q, R, J, Δ, with the following property for all sufficiently large L. Let $R, C \subseteq \Lambda$ be disjoint, $R_0 \subseteq R$, $C_0 \subseteq C$, and suppose $I_\Omega(R:C) \le b$. If*

$$\mathrm{dist}(R_0, \Lambda\setminus R) > w, \qquad \mathrm{dist}(C_0, \Lambda\setminus C) > w,$$
$$w = C_p(1 + b + \log L),$$

then $I_\Omega(R_0 : C_0) \le L^{-p}$.

Proof. If either core is empty there is nothing to prove. Let $\Psi = \Omega\otimes\Omega$ on two replicas, F_R be the unitary exchanging the two replicas at the sites of R, $\Phi = F_R\Psi$, and $Q = \Lambda \setminus (R \cup C)$.

Overlap. Apply Lemma VIII.2 to $(T, U, E) = (R, Q, C)$ to obtain an isometry $V : \mathcal{H}_Q \to \mathcal{B}_R \otimes \mathcal{B}_C$ and unit vectors $a \in \mathcal{H}_R \otimes \mathcal{B}_R$, $a' \in \mathcal{H}_C \otimes \mathcal{B}_C$ satisfying $|\langle a \otimes a', \widetilde{\Omega}\rangle| \ge \mathrm{e}^{-b/2}$, where $\widetilde{\Omega} = (\mathrm{I}\otimes V)\Omega$. The largest Schmidt probability (marginal eigenvalue) of $\widetilde{\Omega}$ across the bipartition $R\mathcal{B}_R \,|\, C\mathcal{B}_C$ is therefore at least e^{-b}, so $\mathrm{Tr}\,\rho^2_{\widetilde{\Omega}, R\mathcal{B}_R} \ge \mathrm{e}^{-2b}$. On the two replicas of Q, define the Hermitian contraction $W = (V^\dagger \otimes V^\dagger) F_{\mathcal{B}_R} (V \otimes V)$, where $F_{\mathcal{B}_R}$ swaps the two copies of $\mathcal{B}_R$. Since V acts outside R, the replica-swap identity $\langle \xi^{\otimes 2}, F_A \xi^{\otimes 2}\rangle = \mathrm{Tr}\,\rho^2_{\xi,A}$ gives

$$\begin{aligned} z := \langle \Phi, W\Psi\rangle &= \langle \widetilde{\Omega}^{\otimes 2}, F_R F_{\mathcal{B}_R} \widetilde{\Omega}^{\otimes 2}\rangle \\ &= \mathrm{Tr}\,\rho^2_{\widetilde{\Omega}, R\mathcal{B}_R} \;\ge\; \mathrm{e}^{-2b}. \end{aligned} \tag{25}$$

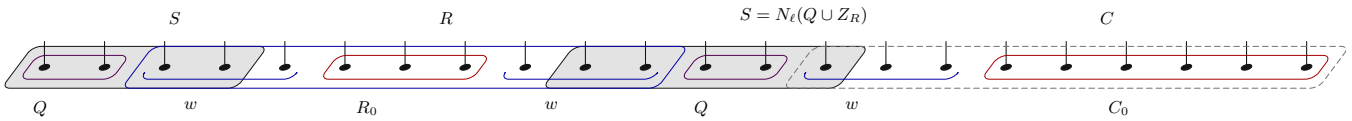


FIG. 15. The reset geometry, drawn in one replica of $\Psi = \Omega \otimes \Omega$: the cores $R_0 \subset R$ and $C_0 \subset C$, the buffer $Q = \Lambda \setminus (R \cup C)$, the margins of width w, the set Z_R near the cut edges of R, and the support $S = N_\ell(Q \cup Z_R)$ of the unitary, which stays clear of both cores.

Two-sided Gaussian filtering. Let $\mathbb{H} = H \otimes \mathrm{I} + \mathrm{I} \otimes H$ and $\mathbb{H}' = F_R \mathbb{H} F_R$. Both are range-$R$ Hamiltonians on the doubled chain with uniformly bounded interaction sums, unique ground vectors Ψ and Φ, common ground energy $\mathcal{E} = 2E_0$, and spectral gap at least Δ. Their difference $\mathbb{H}' - \mathbb{H}$ is a sum of at most $N_R|\partial R|$ terms of norm at most $2J$, namely those terms of $\mathbb{H}$ whose support meets both R and $\Lambda \setminus R$ (terms supported entirely inside R are merely exchanged between the two identical replicas); let Z_R be the union of their supports, which lies within distance R of the cut edges of R. Put $B_* = 1 + b + \log L$, $r_* = 2p + 10$, and $h = AB_*$ with $A\Delta^2/2 \geq r_* + 30$, and define

$$g_h(t) = (2\pi h)^{-1/2} \mathrm{e}^{-t^2/(2h)},$$
$$M = \int_{\mathbb{R}} g_h(t)\, \mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}}\, dt.$$

The operator M is a contraction, and by the spectral theorem $M\Psi = \mathrm{e}^{-h(\mathbb{H}'-\mathcal{E})^2/2} W\Psi$ and $M^\dagger \Phi = \mathrm{e}^{-h(\mathbb{H}-\mathcal{E})^2/2} W^\dagger \Phi$. Because $\mathbb{H}'$ and $\mathbb{H}$ have gap at least Δ above $\mathcal{E}$,

$$\|M\Psi - z\Phi\|,\ \|M^\dagger \Phi - z\Psi\| \leq \mathrm{e}^{-h\Delta^2/2} \leq \mathrm{e}^{-10b} L^{-r_*}. \quad (26)$$

Truncating the time integral to $|t| \leq T = T_0 B_*$ incurs an error of at most $2\mathrm{e}^{-T_0^2 B_*/(2A)}$, which is likewise at most $\mathrm{e}^{-10b} L^{-r_*}$ once $T_0^2/(2A) \geq r_* + 30$.

Localisation. Factor the integrand as $\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}} = (\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'}) V(t)$ with $V(t) = \mathrm{e}^{it\mathbb{H}'} \mathrm{e}^{-it\mathbb{H}}$, whose left generator $G(t) = \mathrm{e}^{it\mathbb{H}'}(\mathbb{H}' - \mathbb{H})\mathrm{e}^{-it\mathbb{H}'}$ is a sum of the boundedly many terms supported in Z_R. Applying Lemma VIII.3 to W (supported in Q, of norm at most 1) and to each term of $G(t)$ yields Hermitian localised approximants $W_\ell(t) = \mathcal{P}_{N_\ell(Q)}\big(\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'}\big)$ and $G_\ell(t) = \mathcal{P}_{N_\ell(Z_R)} G(t)$ satisfying

$$\|\mathrm{e}^{it\mathbb{H}'} W \mathrm{e}^{-it\mathbb{H}'} - W_\ell(t)\| + \|G(t) - G_\ell(t)\| \leq CL^2\, \mathrm{e}^{v|t| - \mu\ell},$$

where the factor L^2 bounds $|Q|^2$ and $|\partial R|$. Let $V_\ell(t)$ be the unitary evolution generated by $G_\ell(t)$, which is supported in $N_\ell(Z_R)$; Duhamel's formula gives $\|V(t) - V_\ell(t)\| \leq CL^2|t|\mathrm{e}^{v|t|-\mu\ell}$. Consequently, $W_\ell(t)V_\ell(t)$ is a contraction supported in $S := N_\ell(Q \cup Z_R)$ that approximates the integrand uniformly for $|t| \leq T$ with error at most $CL^2(1+T)\mathrm{e}^{vT-\mu\ell}$. Choose $\ell = \lceil K_{\mathrm{loc}} B_* \rceil$ with K_{loc} fixed large enough (after T_0) that this error is at most $\mathrm{e}^{-10b} L^{-r_*}$, using $\log L \leq B_*$ and $\log(1+T) \leq C + B_*$. Averaging the localised integrands against the truncated Gaussian yields a contraction M_{loc} supported in S with

$$\|M_{\mathrm{loc}}\Psi - z\Phi\|,\ \|M_{\mathrm{loc}}^\dagger \Phi - z\Psi\| \leq C\mathrm{e}^{-10b} L^{-r_*}. \quad (27)$$

Polar decomposition. Rescale by setting $D = M_{\mathrm{loc}}/z$. Since $z \geq \mathrm{e}^{-2b}$, we have $\|D\| \leq \mathrm{e}^{2b}$ and $\|D\Psi - \Phi\|, \|D^\dagger\Phi - \Psi\| \leq C\mathrm{e}^{-8b} L^{-r_*}$. Writing $(D^\dagger D - \mathrm{I})\Psi = D^\dagger(D\Psi - \Phi) + (D^\dagger\Phi - \Psi)$ then gives $\|(D^\dagger D - \mathrm{I})\Psi\| \leq C\mathrm{e}^{-6b} L^{-r_*}$. Because D acts on the finite-dimensional space $\mathcal{H}_S$, its polar partial isometry extends to a unitary U on $\mathcal{H}_S$ tensored with the identity outside S. Using the scalar inequality $(\sqrt{x} - 1)^2 \leq (x-1)^2$ for $x \geq 0$, we obtain $\|(U - D)\Psi\| = \|(\mathrm{I} - |D|)\Psi\| \leq \|(\mathrm{I} - D^\dagger D)\Psi\|$, and therefore

$$\|U\Psi - \Phi\| \leq C\mathrm{e}^{-6b} L^{-r_*} \leq CL^{-r_*}. \quad (28)$$

Support and conclusion. By hypothesis, every site of Q lies at distance greater than w from both R_0 and C_0. Any interaction term whose support meets both R and $\Lambda \setminus R$ contains a site $y \notin R$ (at distance greater than w from R_0) and a site $x \in R \subseteq \Lambda \setminus C$ (at distance greater than w from C_0); since its support has diameter at most R, every site of Z_R lies at distance greater than $w - R$ from both cores. Choosing $C_p > K_{\mathrm{loc}} + R + 3$ ensures that $S = N_\ell(Q \cup Z_R)$ is disjoint from $R_0 \cup C_0$ in both replicas. Thus U leaves the two-replica marginal on $R_0 C_0$ unchanged. In $\Phi = F_R\Psi$, the marginal on R_0C_0 of the *first* replica is $\rho_{R_0} \otimes \rho_{C_0}$, because R_0 is swapped in from the second replica whereas C_0 remains in the first. By (28) and the contractivity of the partial trace, the ground-state marginal $\rho_{R_0C_0}$ lies within trace distance CL^{-r_*} of $\rho_{R_0} \otimes \rho_{C_0}$. Applying the Alicki–Fannes–Winter continuity bound $|S(A|B)_\rho - S(A|B)_\sigma| \leq C\sqrt{\delta}\log(\mathrm{e}\dim\mathcal{H}_A)$ with $\delta = \frac{1}{2}\|\rho - \sigma\|_1$ [6] to $I(R_0 : C_0) = S(R_0) - S(R_0|C_0)$, and using $\log\dim\mathcal{H}_{R_0} \leq L\log q$, gives $I_\Omega(R_0 : C_0) \leq CL^{1-r_*/2} = CL^{-p-4} \leq L^{-p}$ for all sufficiently large L. □

C. The one-dimensional separation lemma

In words. Combining the one-dimensional area law with the information reset yields polynomial decoupling at logarithmic separation. By Theorem I.1, any interval R (or union of k_0 intervals) has entanglement entropy at most $k_0 C_{\mathrm{int}}$, so its mutual information with any disjoint region C is bounded by the constant $2k_0 C_{\mathrm{int}}$. Feeding this uniform bound into Lemma VIII.4 shows that any union T of at most k_0 intervals and any set E separated from T by a distance of order $\log L$ satisfy $I_\Omega(T : E) \leq L^{-p}$. Because the one-dimensional area law bounds interval entropies uniformly at every scale, this

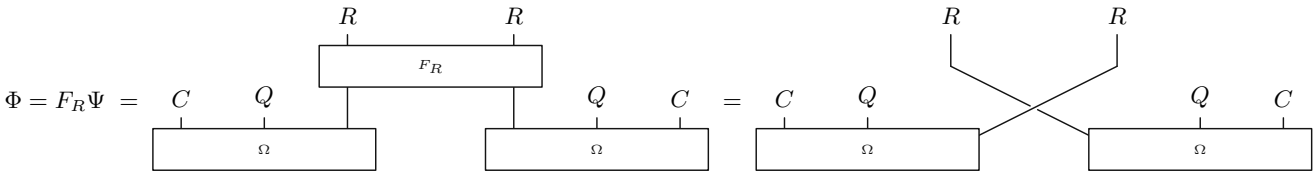


FIG. 16. The swap F_R exchanges the two copies of Ω on R only (the first copy is drawn mirrored so that the two R legs are adjacent). In $\Phi = F_R\Psi$ the core R_0 of the first copy comes from the second copy while C_0 does not, so the first copy's marginal on R_0C_0 is $\rho_{R_0} \otimes \rho_{C_0}$.

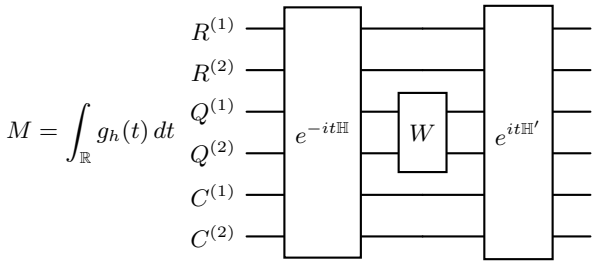


FIG. 17. The two-sided Gaussian filter: evolve with the doubled Hamiltonian, apply W on the two copies of the buffer, evolve back with the swapped Hamiltonian, and average against a Gaussian in time. It sends $W\Psi$ to a multiple of Φ and $W^\dagger\Phi$ to a multiple of Ψ, up to $e^{-h\Delta^2/2}$.

single step establishes logarithmic decoupling for intervals of arbitrary length, from a few sites up to the entire chain. This replaces the entire Section 3 of the two-dimensional paper (coarse collar, cell conditional information, angular induction).

Precisely.

Lemma VIII.5 (Information at logarithmic separation)**.** *For every $p > 0$ and $k_0 \geq 1$ there is $D_p = D_p(q, R, J, \Delta, k_0)$ such that for all sufficiently large L, every $T \subseteq \Lambda$ that is a union of at most k_0 intervals, and every $E \subseteq \Lambda$ with* $\operatorname{dist}(T, E) \geq D_p \log L$*,*

$$I_\Omega(T : E) \leq L^{-p}.$$

Proof. Let $b = 4k_0C_{\mathrm{int}}$, let $w = C_p(1 + b + \log L)$ be the margin width from Lemma VIII.4, and choose D_p so that $D_p \log L \geq 3w+3$. Put $R = T^{+(w+1)}$ (a union of at most k_0 intervals), $C = \Lambda \setminus T^{+(2w+2)}$, $R_0 = T$, and $C_0 = E$. Then $\operatorname{dist}(R_0, \Lambda\setminus R) \geq w+1$ and $\operatorname{dist}(C_0, \Lambda\setminus C) \geq w+1$. Moreover, $I(R : C) \leq 2S(R) \leq 2k_0C_{\mathrm{int}} \leq b$ by Theorem I.1, using subadditivity across the at most k_0 intervals of R together with $I(R : C) = S(R) + S(C) - S(RC) \leq 2S(R)$ from the Araki–Lieb inequality. Applying Lemma VIII.4 gives $I_\Omega(T : E) \leq L^{-p}$. □

Remark VIII.6 (A route without Lieb–Robinson bounds)**.** The information-reset lemma is the only place in Part II where a Lieb–Robinson bound is invoked. It can be replaced by the algebraic amplification argument of [1, Section 10], which uses exact positive quasi-local constraints with stretched-exponential tails e^{-cl^α} ($\alpha = p/(p+1)$), random products of the associated contractions, and a polar decomposition, without requiring a propagation bound for the unitary dynamics. With the area law as input, that argument yields $I(T : E) \leq L^{-M}$ whenever E lies beyond distance $C(\log L)^{1/\alpha}$ of an interval T. Running the protocol below on blocks of that width produces a matrix product state of bond dimension $\exp(O((\log L)^{1/\alpha}))$, which is quasi-polynomial for every fixed p and approaches polynomial only in the large-p limit. Achieving strictly polynomial bond dimension through the protocol route relies on the logarithmic separation provided by Lemma VIII.5.

IX. DISTRIBUTED CONSTRUCTIONS AND THE THREE MOVES

In words. To convert logarithmic decoupling into a tensor network, we track how ownership of the chain's site registers is transferred among a finite set of *parties*. Each party holds a collection of registers (tensor factors), which may include complete copies of the ground vector prepared privately. A *gate* is a linear contraction applied to the registers of a small number of parties. A gate is *private* if it acts on a single party's registers; private gates are free. A non-private gate is charged through only two combinatorial quantities: how many parties it involves, and how many non-private gates each party participates in over the entire protocol. In addition, every non-private gate must admit a *polynomial expansion*: it must be a sum of polynomially many terms, each a composition of private contractions, preparations of normalised bipartite vectors shared by two parties, contractions against normalised bipartite covectors shared by two parties, and transfers of polynomial-dimensional registers. Crucially,

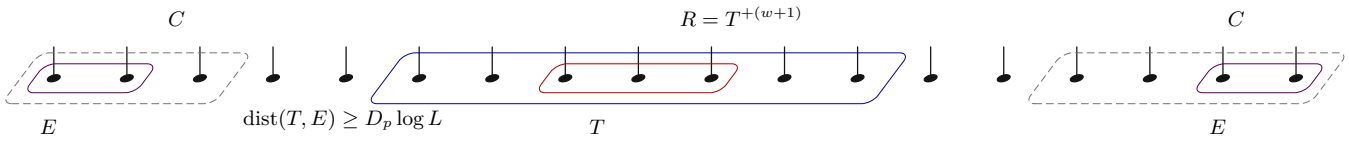


FIG. 18. The separation lemma: an interval T, its enlargement R, the far region C and the set $E \subseteq C$ at distance $D_p \log L$ from T. The area law bounds $I(R:C)$ by a constant and the reset makes $I(T:E) \le L^{-p}$.

the Hilbert-space dimensions of the private registers and of the shared bipartite vectors are unrestricted.

Separating private Hilbert-space dimension from gate participation count is the key mechanism of the protocol: as shown in Section XI, the final physical density matrix of any such distributed construction can be compressed into a tensor network on the party graph with polynomial bond dimension, regardless of the dimensions of the private registers or shared resources. One party may therefore prepare the exact ground vector Ω privately, and pairs of parties may share arbitrarily entangled bipartite states, provided each party takes part in only a bounded number of non-private gates.

A *sheet* is a complete set of chain registers $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$ equipped with an assignment of a party (its owner) to each site; its reference vector is a copy of Ω placed on those registers in canonical site coordinates (changing ownership relabels which party holds a site register without permuting sites). Three elementary moves modify the ownership assignment of one or two sheets while keeping the actual state vector within $O(L^{-p/2})$ of the tensor product of the reference vectors. A *birth* hands a set T of sites from a party $P_\circ$ to another party $Q_\circ$ when T carries little mutual information with the sites $P_\circ$ does not own; a *death* hands T back when the same condition holds after the move; and a *two-sheet exchange* swaps the two ownership assignments inside an interval Y under the analogous information condition, without moving any register between parties.

Precisely. A *distributed construction* of size parameter L consists of a finite set of parties, each initialised with a finite-dimensional register space in a normalised product state, together with a sequence of gates, each a linear contraction between specified input and output register spaces of the participating parties while all other registers act as spectators. Registers to be discarded are retained until a final partial trace. An *allowed monomial* is a composition of private contractions, preparations of normalised vectors shared between two participating parties, contractions against normalised covectors shared between two participating parties, and identity transfers of registers of uniformly polynomial dimension. A non-private gate is *admissible* if it is a sum of uniformly polynomially many allowed monomials with uniformly polynomially bounded scalar coefficients, each monomial containing a bounded number of shared vectors, shared covectors, and transfers; a contraction within L^{-a} of such a sum, for a fixed $a > 0$, is also admissible after rescaling by $1+L^{-a}$. A *bounded change* is a sequence of a bounded number of admissible gates on a bounded number of parties.

A sheet is $\mathcal{H}_\Lambda = \bigotimes_{x\in\Lambda} \mathbb{C}^q$ with an owner assigned to each site; its reference vector is Ω placed in these registers.

Lemma IX.1 (Birth and death)**.** *In a sheet, let T be a set of sites owned by $P_\circ$, and let $E = \{x :$ the owner of x is not $P_\circ\}$ and $U = \Lambda \setminus (T \cup E)$. If $I_\Omega(T:E) \le L^{-2p}$, then changing the owner of T from $P_\circ$ to any party $Q_\circ$ (a* birth*) is a bounded change involving $P_\circ$ and $Q_\circ$ only, consisting of private maps and one shared normalised vector, with reference-vector error at most L^{-p}. The reverse change (a* death*, returning T from $Q_\circ$ to $P_\circ$) is a bounded change with the same error whenever the ownership and information conditions hold in the sheet* after *the death; it uses one shared normalised covector instead.*

Proof. Let V, s, s' be given by Lemma VIII.2 for (T, U, E), and put $\mathcal{V} = \mathrm{I}_{TE} \otimes V$ and $\Pi = |s\rangle\langle s|_{T\mathcal{B}_T} \otimes \mathrm{I}_{E\mathcal{B}_E}$. Before the birth, all of $T \cup U$ is owned by $P_\circ$. At $P_\circ$, apply V on U; contract $T\mathcal{B}_T$ privately against $\langle s|$ (both registers are held by $P_\circ$); supply the shared vector s on $T\mathcal{B}_T$ with T now owned by $Q_\circ$ and $\mathcal{B}_T$ retained by $P_\circ$; and apply $V^\dagger$ at $P_\circ$. The composite map is the Hermitian contraction $B = \mathcal{V}^\dagger \Pi \mathcal{V}$, which satisfies $\|(B - \mathrm{I})\Omega\| \le \|(\Pi - \mathrm{I})\mathcal{V}\Omega\| = \|(\Pi - \mathrm{I})(\mathcal{V}\Omega - s \otimes s')\| \le L^{-p}$. For a death, apply V at $P_\circ$, contract $T\mathcal{B}_T$ using the shared covector $\langle s|$ held across $Q_\circ$ and $P_\circ$, re-insert s privately at $P_\circ$, and apply $V^\dagger$: this implements the same operator B with reversed ownership. □

Lemma IX.2 (Two-sheet exchange)**.** *Consider two sheets and an interval $Y \subseteq \Lambda$. Inside Y, exchange the two ownership assignments (sheet 1 at $x \in Y$ receives the owner sheet 2 had, and conversely); outside Y, keep both assignments unchanged. Let $P_\circ$ be a party, and set*

$$Z = \{x : \text{the two owners of } x \text{ are not both } P_\circ\},$$
$$T = Z \cap Y, \quad E = Z \setminus Y, \quad U = \Lambda \setminus Z.$$

If $I_\Omega(T:E) \le L^{-2p}$, the exchange is implemented by relabelling registers (no register changes party) together with private contractions at $P_\circ$, with error at most $4L^{-p}$ between the tensor products of the old and new reference vectors.

Proof. Let $\mathcal{R}$ relabel the two sheets inside Y: this is a private identification of tensor factors because the new owner of sheet 1 at $x \in Y$ coincides with the old owner of sheet 2 at x. In canonical site coordinates, $\mathcal{R}$ acts as the

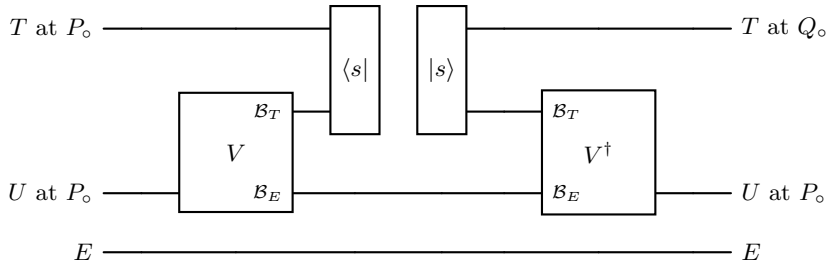


FIG. 19. The birth circuit, read left to right: V splits the buffer, $\langle s|$ closes T and its purifier, a fresh pair s re-emits them with T held by the new owner, and $V^\dagger$ recombines.

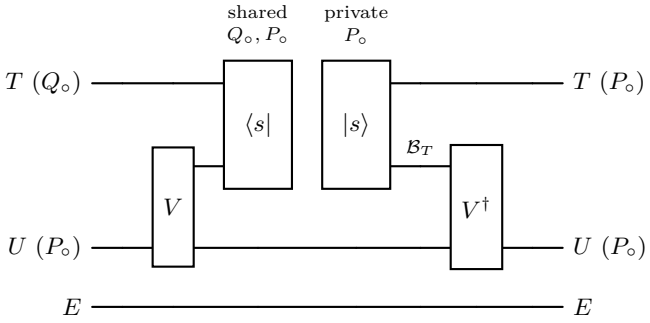


FIG. 20. The death circuit: the same operator with the ownership reversed, using the shared covector $\langle s|$ across the two owners and a private re-insertion of s.

swap F_Y exchanging the two sheets on Y. All registers of U on both sheets are held by $P_\circ$ both before and after $\mathcal{R}$. Let V, s, s' be given by Lemma VIII.2 for (T, U, E), put $\mathcal{V} = \mathrm{I}_{TE} \otimes V$, and define $D_U = (V^{\otimes 2})^\dagger F_{\mathcal{B}_T} V^{\otimes 2}$ on the two copies of U; this is a private contraction at $P_\circ$ obtained by applying V to both copies, swapping the two $\mathcal{B}_T$ factors, and applying $V^\dagger \otimes V^\dagger$. Writing $A = Y \cap U$, apply the private swap F_A after $\mathcal{R}$ and then apply D_U. Since $Y = T \sqcup A$, the net action on the two copies in canonical coordinates is

$$D_U F_A F_Y = D_U F_T = (\mathcal{V}^{\otimes 2})^\dagger F_{T\mathcal{B}_T} \mathcal{V}^{\otimes 2},$$

because F_T acts outside U and commutes with $V^{\otimes 2}$. Put $\zeta = s \otimes s'$ and $\delta = \|\mathcal{V}\Omega - \zeta\| \le L^{-p}$. Then $\|(\mathcal{V}\Omega)^{\otimes 2} - \zeta^{\otimes 2}\| \le 2\delta$, $\zeta^{\otimes 2}$ is invariant under $F_{T\mathcal{B}_T}$ (which swaps two identical factors s), and $(\mathcal{V}^{\otimes 2})^\dagger \mathcal{V}^{\otimes 2} = \mathrm{I}$, so

$$\begin{aligned}&\big\|(\mathcal{V}^{\otimes 2})^\dagger F_{T\mathcal{B}_T} \mathcal{V}^{\otimes 2} \Omega^{\otimes 2} - \Omega^{\otimes 2}\big\| \\ &\qquad \le \|(F_{T\mathcal{B}_T} - \mathrm{I})\big((\mathcal{V}\Omega)^{\otimes 2} - \zeta^{\otimes 2}\big)\| \le 4\delta.\end{aligned}$$

In canonical coordinates both the old and new two-sheet reference vectors equal $\Omega \otimes \Omega$ (since sheet registers are relabelled rather than moved), completing the proof. □

Remark IX.3. In the two-dimensional source [2], the analogous ownership moves carry encoded holes and tags at trijunctions where three owners meet. On a line, the closures of intervals with distinct owners meet only at two-party interfaces, so no trijunctions occur and no boundary encodings are required. This is the first of two reasons why the adaptive patch constraints of [2, Section 4] are unnecessary in one dimension.

X. THE SWEEP PROTOCOL

In words. We now schedule the ownership moves across the chain. Divide Λ into consecutive blocks of $t = 20w$ sites, where $w = \lceil D_{24} \log L \rceil$ is the logarithmic decoupling distance of Lemma VIII.5; the final block absorbs the remainder and has length between $3t$ and $4t$. Initially, a single remainder party R_0 owns the entire chain. At the start of step k, the ownership word reads

$$P_1 \,|\, P_2 \,|\, \cdots \,|\, P_k \,|\, R_k,$$

where each party P_i owns block B_i and the remainder party R_k owns the rest of the chain. Two births inside R_k, both permitted because they leave a buffer of width w separating them from all other owners, create the next remainder party R_{k+1} on everything beyond block B_{k+1} (plus a w-buffer) and the next block owner P_{k+1} on the interior of block B_{k+1}. What remains under R_k's ownership are two buffer bands of width w: one wedged between P_k and P_{k+1}, and the other wedged between P_{k+1} and R_{k+1}. Handing these two buffer bands over to their neighbouring owners is the only nontrivial step of the sweep.

Why can a buffer band A wedged between two distinct owners C (on the left) and B (on the right) not simply die into B? Because A touches C directly, it has zero separation from a non-B owner, so the decoupling hypothesis of

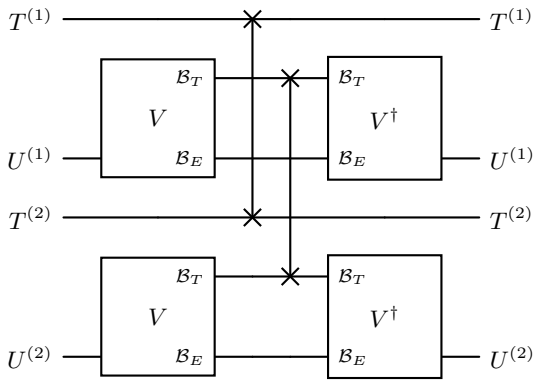


FIG. 21. The exchange correction on the two sheets: after renaming the registers inside Y, the common owner applies V to both buffers, swaps the two purifying factors, and applies the adjoints, which realizes the partial swap on T up to $4L^{-p}$.

Lemma IX.1 is not available. To dissolve the band without ever moving a region near a third party, we use an auxiliary sheet [2] in which the interface is pre-wrapped inside nested buffers of A, B, and C; the sequence transplants verbatim from the edge construction of [2]. In units of the band width w, with the band A occupying $0 < x < 1$, C to its left, and B to its right:

1. Prepare a fresh auxiliary copy of the ground vector held entirely by C. On this auxiliary sheet, birth B on $(-6, -1)$, then birth A on $(-5, -2)$, and then birth C on $(-4, -3)$. The auxiliary ownership word is now $C\,|\,B\,|\,A\,|\,C\,|\,A\,|\,B\,|\,C$.
2. Exchange the main and auxiliary sheets inside the window $Y = (-7, -\frac{7}{2})$. Both endpoints of Y lie inside regions owned by C on both sheets, and all non-C intervals inside Y are separated from all non-C intervals outside Y by a margin of width at least w. After the exchange, the main sheet reads $C\,|\,B\,|\,A\,|\,C\,|\,A\,|\,B$ with interfaces at $-6, -5, -4, 0, 1$, and the auxiliary sheet is retired.
3. On the main sheet, every subsequent move is flanked by a single owner on both sides: the C band $(-4, 0)$ is flanked by A on $(-5, -4)$ and $(0, 1)$, so it dies into A; the merged A band $(-5, 1)$ is flanked by B on $(-6, -5)$ and $(1, 2)$, so it dies into B, reducing the word to $C\,|\,B$ with interface at -6. Finally, birth C on $(-3, 0)$ inside B and let the B band $(-6, -3)$ die into C: the interface returns to $x = 0$ with the original band $(0, 1)$ absorbed into B.

Every move in this sequence modifies an interval separated by distance at least w from all third-party owners, so Lemma VIII.5 supplies the decoupling bound required by Lemma IX.1 or Lemma IX.2. Only the three parties A, B, C participate. Because each party is born once, dissolves the buffer bands on its two sides once, and is otherwise a spectator, every party participates in a uniformly bounded number of gates.

Why use a linear sweep rather than a multiscale dyadic hierarchy? Because the compressed party network must ultimately be routed onto the one-dimensional chain. Each link of the party network connects two parties, and every link crossing a given cut of the chain multiplies the bond dimension across that cut. In a dyadic hierarchy of intervals, every scale contributes a link across the midpoint of the chain (the interval tree has cutwidth $\log L$), which would inflate the bond dimension to $L^{O(\log L)}$. In the linear sweep, every non-private gate links parties associated with blocks at most two apart, so only $O(1)$ links cross any cut of the chain.

Precisely. Fix $p = 12$. Let D_{24} be the constant of Lemma VIII.5 with exponent $2p = 24$ and $k_0 = 1$, and put

$$\begin{aligned} w &= \lceil D_{24} \log L \rceil, \qquad t = 20w, \\ k_* &= \max\{k \geq 0 : \; L - kt \geq 3t\}, \end{aligned} \tag{29}$$

with blocks $B_i = [(i-1)t+1,\, it]$ for $1 \leq i \leq k_*$ and final block $B_{k_*+1} = [k_* t+1, L]$ of length in $[3t, 4t)$. We assume L is large enough that $t \geq 20$ and $k_* \geq 0$ (smaller L are treated directly in Section XII). For a band $[a, a+w)$ with C owning $[a-8w, a)$ and B owning $[a+w, a+2w)$ (and possibly further sites), write $x = (\cdot - a)/w$ for the scaled coordinate.

Construction X.1 (Band dissolution)**.** Suppose the main sheet reads $C\,|\,A\,|\,B$ near a: C owns $x < 0$ down to at least $x = -8$, A owns $0 < x < 1$, and B owns $x > 1$ up to at least $x = 2$, where A, B, C are three distinct parties. Perform the following sequence of moves:

(i) Prepare an auxiliary sheet as a fresh copy of Ω held entirely by C (a private initialisation). On this auxiliary sheet, birth B on $(-6, -1)$, birth A on $(-5, -2)$, and birth C on $(-4, -3)$.

(ii) Exchange the two sheets inside $Y = (-7, -\frac{7}{2})$ with common owner $P_\circ = C$. Retire the auxiliary sheet (its registers remain idle at their current owners).

(iii) On the main sheet, perform the death of the C band $(-4, 0)$ into A, the death of the A band $(-5, 1)$ into B, the birth of C on $(-3, 0)$ inside B, and the death of the B band $(-6, -3)$ into C.

If there is no left owner C (when the band starts at site 1), replace the entire construction by a single death of the band into B.

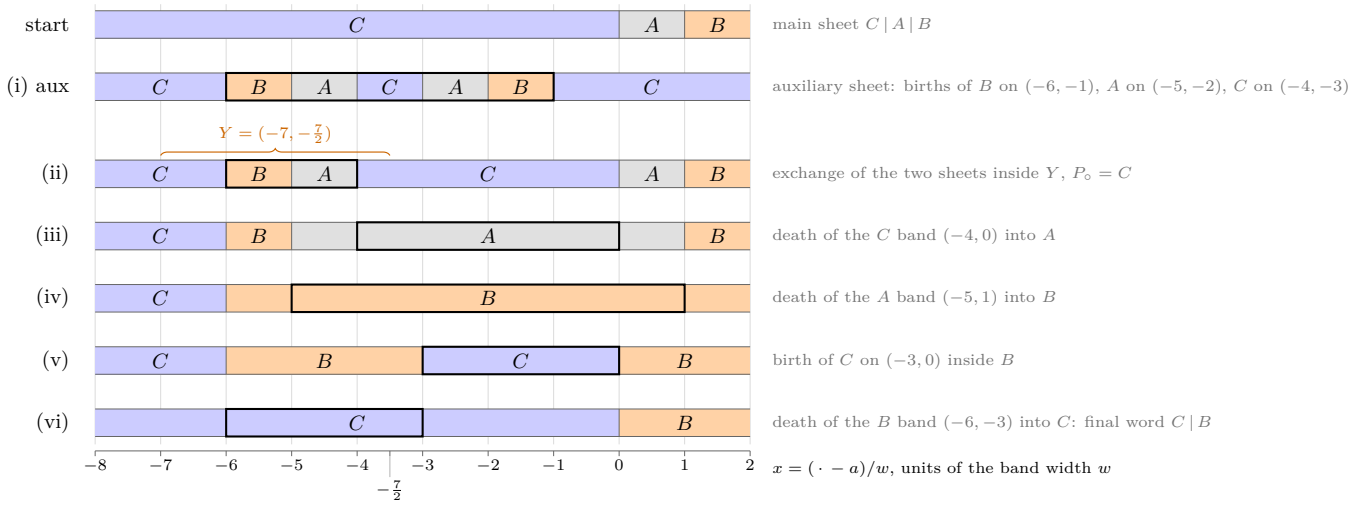


FIG. 22. The band dissolution in units of the band width: the auxiliary word after its three births (the heavier outline marks their union), the exchange window Y, and the sequence of deaths and births that leaves the band to B with the interface back at zero.

Lemma X.2 (The band dissolution is a bounded change)**.** *For all sufficiently large L, each of the eight moves (seven births or deaths and one exchange) in Construction X.1 satisfies the information hypothesis of Lemma IX.1 or Lemma IX.2 with exponent $2p = 24$, so that the construction is a bounded change on the parties A, B, C with total reference-vector error at most $12L^{-12}$. Afterwards, the main sheet reads $C \mid B$ with the interface at $x = 0$, and all other ownership is unchanged.*

Proof. We verify the separation hypothesis for each move; all distances are measured in units of $w \geq D_{24} \log L$, and every modified region is a single interval, so Lemma VIII.5 applies whenever the scaled separation is at least 1. (i) The first auxiliary birth has $E = \varnothing$ and requires no information bound. The second modifies $(-5,-2)$ inside the B band $(-6,-1)$, so the non-B region E lies at distance 1. The third modifies $(-4,-3)$ inside the A band $(-5,-2)$, again at distance 1 from E. The auxiliary word is now $C|B|A|C|A|B|C$ with interfaces at $-6,-5,-4,-3,-2,-1$. (ii) The main sheet is owned by C on $x < 0$, while the auxiliary sheet is owned by C on $x < -6$, $(-4,-3)$, and $x > -1$. Hence $T = Z \cap Y = (-6,-4)$ (owned by B and A on the auxiliary sheet) and $E = Z \setminus Y = (-3,-1) \cup [0,\infty)$, together with $x \leq -8$ if C owns no sites beyond $x = -8$; thus $\operatorname{dist}(T,E) \geq 1$. After the exchange, the main sheet inside Y inherits the auxiliary assignment, yielding $C|B|A|C|A|B$ with interfaces at $-6,-5,-4,0,1$, while the auxiliary sheet is owned by C throughout Y. (iii) On the main sheet, the C band $(-4,0)$ is flanked by A on $(-5,-4) \cup (0,1)$, so the non-A sites after the death lie in $x \leq -5$ and $x \geq 1$, at distance 1. Next, the A band $(-5,1)$ is flanked by B on $(-6,-5) \cup (1,2)$, so the non-B sites after the death lie in $x \leq -6$ and $x \geq 2$, at distance 1. The birth of C on $(-3,0)$ inside B has non-B sites at distance at least 2. Finally, the death of $(-6,-3)$ into C has non-C sites in $x \geq 0$ at distance 3 and, if C owns nothing beyond $x = -8$, in $x \leq -8$ at distance 2. Each of the seven births or deaths incurs error at most L^{-12} and the exchange incurs at most $4L^{-12}$, giving total error at most $11L^{-12} \leq 12L^{-12}$. □

Construction X.3 (The sweep)**.** Initially, a single party R_0 holds the entire sheet. For $k = 0, 1, \ldots, k_* - 1$, let $a = kt + 1$ be the first site of block B_{k+1}, and starting from the ownership word $P_1|\cdots|P_k|R_k$:

(1) Birth R_{k+1} on $[a + t + w, L]$ inside R_k.

(2) Birth P_{k+1} on $[a + w, a + t - 1]$ inside R_k.

(3) Dissolve the band $[a, a + w - 1]$ (owned by R_k) between $C = P_k$ and $B = P_{k+1}$ using Construction X.1; if $k = 0$, let the band die directly into P_1.

(4) Dissolve the band $[a + t, a + t + w - 1]$ (owned by R_k) between $C = P_{k+1}$ and $B = R_{k+1}$ using Construction X.1.

After step $k_* - 1$, set $P_{k_*+1} = R_{k_*}$, which owns the final block B_{k_*+1}.

Proposition X.4 (Properties of the sweep)**.** *For all sufficiently large L, Construction X.3 defines a distributed construction with the following properties:*

1. *There are $2k_* + 1 \leq 2L$ parties. The initial state is a product state across parties: R_0 holds Ω on the main sheet, and every auxiliary copy of Ω is initialised privately at the party C of the corresponding band dissolution.*

2. *There are at most $20k_* \leq 2L$ non-private gates. Each involves at most three parties and is admissible with one shared vector or covector per monomial.*

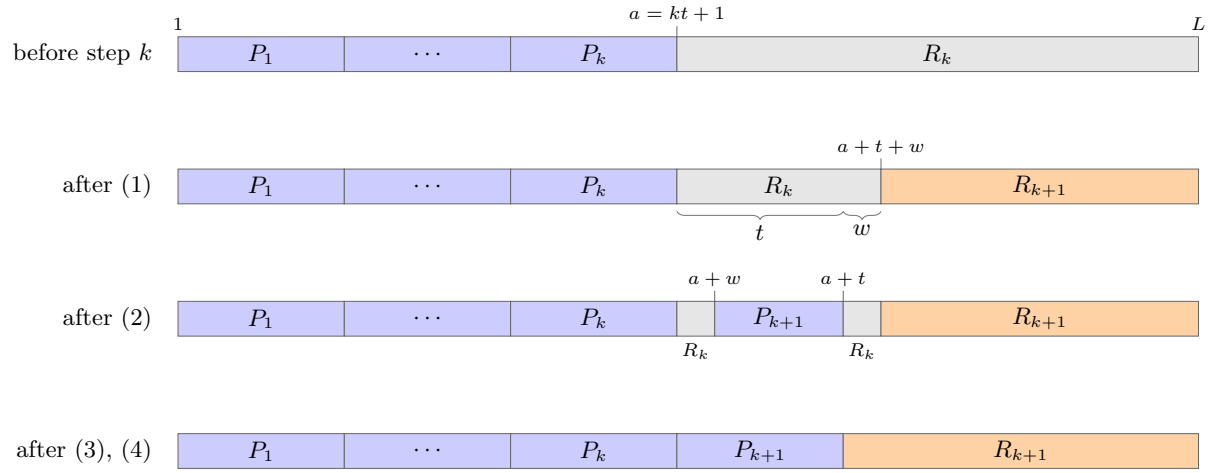


FIG. 23. One step of the sweep (for $k \geq 1$): the word before the step, after the birth of R_{k+1}, after the birth of P_{k+1}, and after the two bands of R_k have been dissolved.

Each party participates in at most 40 *non-private gates over its lifetime.*

3. *The final vector* Ψ *satisfies* $\|\Psi\| \leq 1$ *and* $\|\Psi - \Omega \otimes \Xi\| \leq 26\, L^{-11}$*, where* Ξ *is the tensor product of the retired auxiliary copies of* Ω *in their final registers and* $\|\Xi\| = 1$*. At the end of the sweep, party* P_i *owns precisely the raw registers of block* B_i*, of dimension* $q^{|B_i|} \leq q^{4t} = L^{O(1)}$*, and no other party owns any raw register.*

Proof. At the start of step k, the ownership word is $P_1|\cdots|P_k|R_k$, with R_k owning $[a, L]$ for $a = kt+1$. Move (1) modifies $[a+t+w, L]$ with non-R_k region $E = [1, a-1]$ at distance $t + w$; move (2) modifies $[a + w, a + t - 1]$ with non-R_k region $E = [1, a - 1] \cup [a + t + w, L]$ at distance w. Both moves are valid births by Lemma IX.1 and Lemma VIII.5. Afterwards, R_k owns only the two bands $[a, a + w - 1]$ and $[a + t, a + t + w - 1]$. In (3), $C = P_k$ owns block B_k of length $t = 20w \geq 8w$ to the left of a, and $B = P_{k+1}$ owns $[a+w, a+t-1]$ of length $19w \geq 2w$, so Lemma X.2 applies (when $k = 0$, the direct death of $[1, w]$ into P_1 has non-P_1 region $E = [t+1, L]$ at distance $t - w \geq w$). In (4), $C = P_{k+1}$ owns $[a + w, a + t - 1]$ of length $19w \geq 8w$ and $B = R_{k+1}$ owns $[a + t + w, L]$ of length at least $3t - w \geq 2w$, so Lemma X.2 again applies. Step k concludes with the word $P_1|\cdots|P_{k+1}|R_{k+1}$. The distinct parties are $R_0, \ldots, R_{k_*}$ and $P_1, \ldots, P_{k_*}$, with $P_{k_*+1} = R_{k_*}$. Each step uses at most $2 + 2 \cdot 8 = 18 \leq 20$ moves, giving at most $20k_*$ moves in total, and the final block has length in $[3t, 4t)$.

For participation counts, R_k participates only in step $k - 1$ (as the newborn party in (1) and as B in (4)) and in step k (as $P_\circ$ in (1) and (2), and as A in (3) and (4)); similarly, P_k participates only in step $k - 1$ (born in (2), acting as B in (3) and C in (4)) and in step k (acting as C in (3)). Because each band dissolution consists of seven births or deaths and one private exchange, no party participates in more than $2 \cdot (2 + 2 \cdot 8) = 36 \leq 40$ gates. Every birth or death is a single two-party gate with one shared vector or covector, while every exchange is purely private.

For the error bound, each move maps the current state vector to within the stated error of the updated reference vector, which is the tensor product of Ω on the main sheet and Ω on each active or retired auxiliary sheet in canonical coordinates (Lemmas IX.1 and IX.2; retired auxiliary sheets remain unit-norm spectators). Telescoping the product of contractions bounds the total error by the sum of the per-move errors: two births and two band dissolutions per step contribute at most $(2 + 2 \cdot 12)L^{-12}$, so the cumulative error over k_* steps is at most $26k_*L^{-12} \leq 26L^{-11}$. Since the initial vector has unit norm and every gate is a contraction, $\|\Psi\| \leq 1$. □

Remark X.5 (Why small-patch rewrites are unneeded)**.** In two dimensions [2, Section 4], squares of side length below $K_0 \log L$ must be repainted by "small-patch rewrites" using finite-rank cylinder projections of polynomial rank. In one dimension, the final owners P_i hold blocks of $O(\log L)$ consecutive sites whose Hilbert-space dimension $q^{|B_i|} = q^{O(\log L)}$ is already polynomial in L, and the compression theorem of Section XI accepts polynomial-dimensional physical outputs directly. Factoring each block tensor into single-site matrix-product tensors is then a routine singular value decomposition (Section XII). This is the second reason why the adaptive patch constraints of [2] are unneeded on a chain.

XI. COMPRESSION AND COLUMN EXTRACTION

In words. The sweep protocol is a sequence of linear contractions followed by a partial trace over all registers except the physical block outputs. Its non-private gates still rely

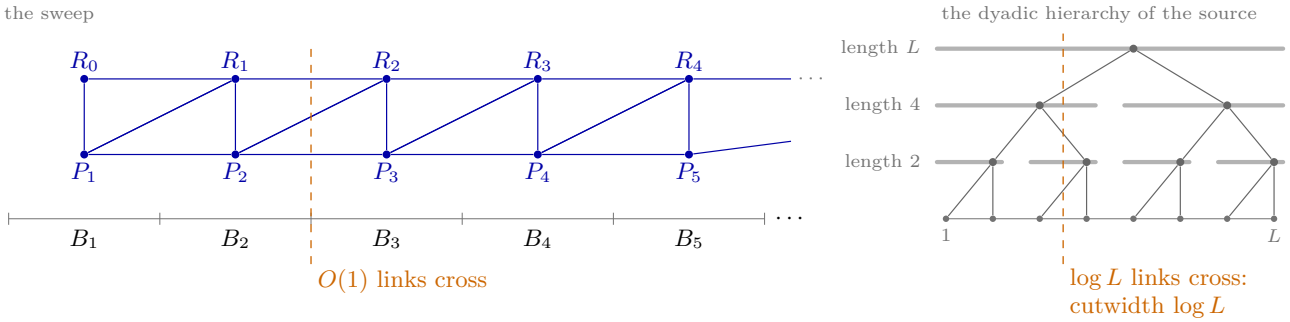


FIG. 24. Left: the party graph of the sweep placed on the chain; a bounded number of links cross any bond. Right: a dyadic hierarchy drawn on a line, where every scale contributes a link across the middle bond.

on shared bipartite vectors of enormous dimension (such as purifications of half-chain marginals). To compress these virtual links to polynomial dimension, the density-compression theorem replaces each shared source operator $|\eta\rangle\langle\tilde{\eta}|$ in the density calculation $\rho = \mathrm{Tr}_{\mathrm{private}} |\Psi\rangle\langle\Psi|$ with the sample average of k independent random product operators $U_j \otimes V_j$ built from its Schmidt decomposition and standard complex Gaussian coefficients. For a polynomial sample size k, the physical output density changes by at most L^{-p} in trace norm. Expanding the error according to the subset of slots where a centred correction $\widehat{X} - X$ acts, each corrected slot involves only its two endpoint parties, and bounded participation ensures that only a bounded number of gates per corrected slot must be expanded into monomials; a Gaussian second-moment computation with Schatten–Hölder then shows that a term with s corrections has expected trace norm at most $k^{-s/2}$ times a fixed polynomial per correction. Because this estimate depends only on the trace-one Schmidt probability weights and the combinatorial participation bounds, it is completely independent of the private Hilbert-space dimensions and of the Schmidt ranks of the shared states. Fixing a deterministic realisation produces an operator tensor network σ with one node per party and polynomial-dimensional links between parties that shared a gate. Extracting a single product-basis column $\sigma|z\rangle$ then converts σ into a ket tensor network with the same virtual links, which approximates the ground vector after normalisation. The theorem and its proof are not geometric and carry over without change; we restate the theorem in the form used here and sketch the estimate.

Precisely.

Theorem XI.1 (Density compression, [2, Theorem 5.2])**.** *Consider a family of distributed constructions indexed by $L \geq 2$ such that: the numbers of parties and gates are uniformly polynomial; the initial vector is a product of normalised private vectors; every gate is a contraction; each non-private gate involves at most b parties and each party participates in at most b non-private gates; each non-private gate is admissible with uniformly polynomial bounds; and the intended physical output of each party has uniformly polynomial dimension, everything else being traced out. Let ρ be the physical output density (positive, trace at most one). For every fixed $p > 0$ there is an operator tensor network σ with one node per party, a bounded number of links at each node, each link joining two parties that participate in a common non-private gate, uniformly polynomial link dimensions, and*

$$\|\sigma - \rho\|_1 \leq L^{-p}.$$

The constants depend only on p, b, and the uniform polynomial bounds, not on private dimensions or on the Schmidt ranks of shared vectors.

Sketch. First eliminate shared covectors: a covector $\langle\eta|$ on a pair of registers is replaced by appending $m-1$ copies of η and averaging over the m cyclic permutations of the m pair registers, forming a contraction T_m with $\|T_m - |\eta\rangle^{\otimes m}\langle\eta|\| = m^{-1/2}$. Attaching a common fixed garbage vector in every branch approximates a gate with K monomials, each containing at most r covectors, to within operator-norm error $rm^{-1/2}\sum|c_\xi|$ by a gate with Km^r monomials using shared vectors only. Choosing m polynomially large makes this error at most $\epsilon/(8M)$ per gate, where M is the number of non-private gates.

Next, in the density calculation, each source operator $X = |\eta\rangle\langle\tilde{\eta}|$ with Schmidt decompositions (λ_a, u_a, v_a) and $(\mu_c, \tilde{u}_c, \tilde{v}_c)$ is replaced by $\widehat{X} = \frac{1}{k}\sum_{j=1}^{k} U_j \otimes V_j$, where $U_j = \sum_{a,c}(\lambda_a\mu_c)^{1/4} g_{ac}^{(j)} |u_a\rangle\langle\tilde{u}_c|$ and $V_j = \sum_{b,d}(\lambda_b\mu_d)^{1/4}\overline{g_{bd}^{(j)}}|v_b\rangle\langle\tilde{v}_d|$ for independent standard complex Gaussians $g_{ac}^{(j)}$. Then $\mathbb{E}\widehat{X} = X$, and by the Gaussian pairing formula the centred coefficients of $\Delta = \widehat{X} - X$ have diagonal covariance $k^{-1}\sqrt{\lambda_a\lambda_b\mu_c\mu_d}$. Expanding the product over all N source slots gives $\widehat{\rho} - \rho' = \sum_{\varnothing\neq S} F_S$, where S indexes the slots carrying a correction Δ. For $|S| = s$, the corrected slots touch at most $2s$ endpoint parties, which participate in at most $2bs$ gates; expand only those gates into monomials and fix the physical basis entries at those parties, at a total weighted cost of at most $Q(L)^s$ for a fixed polynomial Q. For each fixed choice, the network factors into an affected side (holding the corrected endpoints as free inputs) and an exterior side, connected only by the exact shared vectors crossing between them, with Schmidt probability matrices τ and $\tilde{\tau}$. The affected contractions form a block operator O

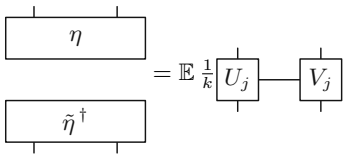


FIG. 25. The random replacement of one pair source: the operator $|\eta\rangle\langle\tilde{\eta}|$ equals the expectation of $\frac{1}{k}\sum_j U_j \otimes V_j$, the two factors joined by the sample index j.

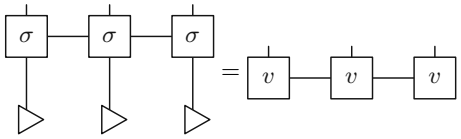


FIG. 26. Column extraction: fixing the input legs of the operator chain by a product-basis covector gives a ket chain with the same bonds.

of norm at most one, the corrected coefficients z_{il} have covariance $k^{-s}\sqrt{p_i\tilde{p}_l}\,\delta\delta$, and the operator fed into the exterior is $Z = \sum_{il} z_{il}\tau^{1/2}O_{li}^{\mathsf{T}}\tilde{\tau}^{1/2}$. Applying the Schatten–Hölder inequality with exponents $(4,2,4)$ and the trace-one weights $\tau,\tilde{\tau}$ gives $\|Z\|_1 \le \|\sum_{il} z_{il}\tilde{\tau}^{1/4}O_{li}\tau^{1/4}\|_2$, whose expected square is $k^{-s}\|\widetilde{R}^{1/4}OR^{1/4}\|_2^2 \le k^{-s}$ with $R = \mathrm{diag}(p_i)\otimes\tau$ and $\widetilde{R} = \mathrm{diag}(\tilde{p}_l)\otimes\tilde{\tau}$. Because the exterior contractions and the physical matrix units are trace-norm contractions, $\mathbb{E}\|F_S\|_1 \le Q(L)^s k^{-s/2}$ and hence $\mathbb{E}\|\widehat{\rho}-\rho'\|_1 \le (1+Q(L)k^{-1/2})^N - 1 \le \epsilon/4$ for a polynomial sample size k. Fixing a realisation of the Gaussian variables yields σ: the monomial index of each gate is shared along a star linking its participating parties, and each slot's sample index $j \in \{1,\dots,k\}$ is shared between its two endpoint parties, while all private contractions, private state preparations, and final private traces are absorbed into each party's local node tensor. □

Lemma XI.2 (Column extraction, [2, Lemma 8.1])**.** *Let Ω be a unit vector on a finite tensor product and σ an operator with $\|\sigma - |\Omega\rangle\langle\Omega|\|_1 \le \eta < 1$. Then some product-basis column $v = \sigma|z\rangle$ is nonzero and satisfies $\min_\theta \|v/\|v\| - \mathrm{e}^{i\theta}\Omega\| \le 2\eta$. Fixing a column in an operator tensor network leaves all virtual dimensions unchanged.*

Proof. Let $E = \sigma - |\Omega\rangle\langle\Omega|$ and $a_z = \langle\Omega|z\rangle$. Since $\sum_z \|E|z\rangle\|^2 = \|E\|_2^2 \le \eta^2$ and $\sum_z |a_z|^2 = 1$, there exists a product-basis label z with $a_z \ne 0$ and $\|E|z\rangle\| \le \eta|a_z|$. For this z, the rescaled column $w = v/a_z$ satisfies $\|w - \Omega\| \le \eta$, so $w \ne 0$ and $\|w/\|w\| - \Omega\| \le 2\eta$. Fixing the bra index at every node contracts only physical legs, leaving all virtual bond dimensions unchanged. □

XII. ASSEMBLY ON THE CHAIN

In words. We now apply density compression and column extraction to the sweep protocol and route the resulting party network along the chain. The parties of the sweep are the block owners P_i, whose physical outputs are the $q^{|B_i|}$-dimensional block registers, and the remainder parties R_k, which have no physical outputs. Every link in the compressed party network connects two parties that shared a non-private gate, and by construction those parties belong to the same or adjacent sweep steps. Placing the node of R_k alongside block B_{k+1} ensures that the chain cut between blocks B_i and B_{i+1} is crossed only by links among the boundedly many parties attached to the four nearest blocks. This bounded congestion implies that each inter-block bond has dimension equal to a bounded product of polynomial link dimensions. Column extraction first converts the compressed operator network into a ket network on the blocks; sequential singular value decompositions across the sites within each block then split each block tensor into single-site tensors whose internal bond dimensions are at most $q^{|B_i|}$ times the inter-block bond dimension, which remains polynomial in L. Chains shorter than the threshold L_0 are represented exactly.

Precisely.

Proof of Theorem I.2. Choose L_0 large enough for Proposition X.4 and for the inequalities below to hold; for $2 \le L < L_0$, every state vector on Λ admits an exact matrix product state representation of bond dimension at most $q^{\lfloor L/2\rfloor} \le q^{L_0/2}$, which is absorbed into the prefactor C. Assume henceforth that $L \ge L_0$.

By Proposition X.4(3) and the inequality $\||u\rangle\langle u| - |v\rangle\langle v|\|_1 \le 2\|u - v\|$ for vectors of norm at most one, the physical output density $\rho = \mathrm{Tr}_{\mathrm{private}}|\Psi\rangle\langle\Psi|$ satisfies $\|\rho - |\Omega\rangle\langle\Omega|\|_1 \le 52L^{-11}$. The sweep satisfies the hypotheses of Theorem XI.1 with participation bound $b = 40$ and physical output dimensions at most $q^{4t} \le L^{80D_{24}\log q+1}$. Applying Theorem XI.1 with target accuracy L^{-5} yields an operator tensor network σ on the party graph with $\|\sigma - |\Omega\rangle\langle\Omega|\|_1 \le L^{-5} + 52L^{-11} \le 2L^{-5}$, bounded node degree, and link dimensions at most $C_1 L^r$. By Lemma XI.2, some product-basis column of σ is a nonzero ket tensor network v with the same virtual links satisfying $\min_\theta \|v/\|v\| - \mathrm{e}^{i\theta}\Omega\| \le 4L^{-5} \le L^{-1}$.

Placement and routing. Assign P_i to block B_i and R_k to block B_{k+1}, and contract the two party nodes assigned to each block into a single block node (links between them cross no inter-block cut). By the proof of Proposition X.4, any two parties sharing a gate belong to $\{P_k, P_{k+1}, R_k, R_{k+1}\}$ for some step k, and are therefore assigned to the same or adjacent blocks (our bound below accommodates blocks up to two apart). The cut between blocks B_i and B_{i+1} is thus crossed only by links incident to parties assigned to blocks $B_{i-1},\dots,B_{i+2}$. There are at most 8 such parties, each of degree at most $5 \cdot 40$ (at most two star links and three sample links per three-party gate), so at most $\chi := 1600$ links cross any inter-block cut. Route each link along the chain between its endpoint blocks and bundle the links crossing each inter-block cut into a single bond index of dimension at most $(C_1 L^r)^\chi$. The tensor at block

B_i is then a linear map from its left and right inter-block bond spaces to the block's physical Hilbert space $\mathcal{H}_{B_i}$ of dimension at most q^{4t}. Writing $B_i = \{x_1 < \cdots < x_n\}$, perform successive singular value decompositions across the internal cuts $x_1|x_2\cdots x_n$, $x_1x_2|x_3\cdots x_n$, …, treating the left and right inter-block bond indices as external legs attached to x_1 and x_n. The resulting single-site tensors have internal bond dimensions at most $\min\{q^j(C_1L^r)^\chi,\ q^{n-j}(C_1L^r)^\chi\} \le q^{2t}(C_1L^r)^\chi$, while the inter-block bonds remain unchanged. Thus v is a matrix product state on Λ with open boundary conditions and bond dimension at most $q^{2t}(C_1L^r)^\chi \le CL^c$, with exponent $c = 40D_{24}\log q + r\chi$. □

Remark XII.1 (Order of constants)**.** Fix q, R, J, Δ. Theorem I.1 determines the interval entropy bound C_{int}; Lemma VIII.4 at exponent $p = 24$ determines C_{24}, which fixes D_{24} and the block length $t = 20w$; the party and gate counts together with the participation bound $b = 40$ are then fixed; Theorem XI.1 at accuracy L^{-5} fixes the covector permutation count m, the Gaussian sample size k, and the link-dimension exponent r; and the congestion bound $\chi = 1600$ is an absolute numerical constant. None of these constants depends on the specific Hamiltonian H or on the chain length L. Over the full sweep, the protocol prepares $O(L/\log L)$ auxiliary copies of Ω, all held privately.

XIII. DISCUSSION

In words. *Comparison with earlier one-dimensional proofs.* Hastings' proof of the one-dimensional area law [3] shows that the half-chain marginal has an entanglement spectrum whose tail decays exponentially beyond rank $\exp(O(1))$ by combining a Lieb–Robinson bound, a spectral filter that approximately projects onto the ground state, and a bound on the approximate factorisation of the filtered state. Arad, Kitaev, Landau, and Vazirani [4] replace the time-evolution filter with an approximate ground-state projector constructed from Chebyshev polynomials of controlled Schmidt rank, which is then amplified by bootstrapping. By contrast, the proof of Theorem I.1 in Part I uses no real-time dynamics at all: the spectral gap enters only to bound the overlap of a variationally tilted or filtered vector with the ground vector, and the entanglement entropy is extracted directly from operator norms via Hölder's inequality, a three-lines interpolation, and the Araki–Lieb triangle inequality, yielding a linear $O(J/\Delta)$ scaling in the inverse gap. For Theorem I.2, this note presents two distinct constructions of the matrix product state: the direct two-line route of Section VII, which combines the marginal tail bound (Lemma III.1) with the area law (Theorem I.1) and is likewise completely free of dynamics, and the distributed protocol route of Part II, which invokes a Lieb–Robinson bound in a single place (the information reset, Lemma VIII.4) but in a way that has no analogue in Hastings' construction (a two-replica swap undone by a local unitary), and otherwise relies solely on Uhlmann-type ownership moves, random Gaussian compression, and product-basis column extraction.

Simplifications specific to one dimension. Three geometric features of the line simplify the two-dimensional arguments of [1, 2], each used in exactly one place. First, the shell around any interval is crossed by $O(1)$ interaction terms, so the buffered discount (Lemma IV.2) costs a constant and needs only a constant-width buffer. Second, no three interval owners ever meet at a single point on a line, eliminating trijunctions and the associated boundary encodings in the ownership moves (Section IX). Third, the chain can be swept sequentially from left to right, giving a party graph of $O(1)$ cutwidth rather than the $\log L$ cutwidth of a dyadic tree (Section X). Everything else in [1, 2] is either unchanged or disappears.

Why the two-dimensional obstructions are genuine. None of these three one-dimensional shortcuts extends back to two dimensions. In two dimensions, the Araki–Lieb closure fails because the buffer surrounding a region is a thin annulus whose own entanglement entropy is a priori unknown; a linear sweep fails because repainting a two-dimensional region exposes a one-dimensional boundary along which a third owner is adjacent, necessitating the edge-band dissolutions and trijunction treatments of [2]; and sequential Schmidt truncation across nested cuts does not produce a two-dimensional tensor network on the lattice. The multiscale machinery of [1, 2] is therefore dictated by genuine two-dimensional geometric and topological obstructions that simply do not arise on a chain.

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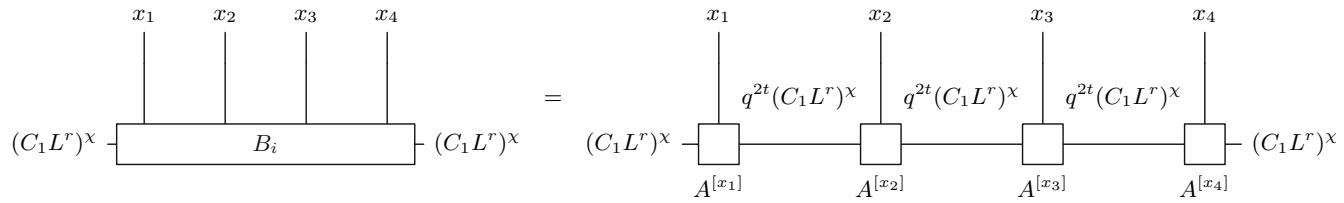


FIG. 27. A block tensor with its two block bonds and t physical legs is split into t site tensors by successive singular value decompositions; the new bonds are at most q^{2t} times the block-bond dimension.